Cross-Assessment of CRISPR Delivery and Editing Safety: Risk Modeling for Clinical Translation

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Gene-editing delivery and safety evidence is scattered across virology, nuclease protein engineering, and clinical translation literatures that rarely cite one another, leaving the field without an integrated, queryable account of how delivery choices propagate into safety risk. This review addresses that gap by constructing a structured evidence base over the CRISPR delivery-safety literature and interrogating it through five linked research questions spanning delivery-method attribution, causal risk chains, benchmark comparability, limitation structure, and cross-domain metric transfer. The evidence base codes each study along seven dimensions—limitation, problem, method, paper, dataset, metric, and assumption—and records cross-work links, enabling both explicit and implicit associations to be recovered and audited. Across the reviewed evidence, the four delivery routes the four delivery routes—viral vectors, electroporation, biomaterials-mediated delivery, and RNP delivery—do not bind distinct safety problems but converge on the same core nodes of off-target effects and immunogenicity, differing instead in association strength and in whether they cause or mitigate harm. Causal evidence linking vector immunogenicity to editing persistence and clinical risk is quantitatively established yet fragmented across three disjoint discourse systems, and roughly 60% of limitations concentrate in the validation/deployment stage under evidence-value uncertainty. Benchmark topology is star-shaped and radially dispersed, with almost no shared metric semantics across assays, so no genuine cross-benchmark normalization anchor exists. Transfer analysis from LLM safety metrics yields a three-tier gradient: uncertainty dichotomies and resampling confidence intervals transfer near-losslessly, attack success rate requires semantic substitution with polarity inversion, and the principle that aggregate scores are comparative rather than universal thresholds transfers most powerfully—precisely by exposing that the field has yet to establish any consensus threshold to critique. The review's central claim is that delivery safety is better modeled as a shared, parameterized risk surface than as method-specific hazard lists, and that the field's missing integration is a shared evidence base rather than another method-specific review.

Additional Key Words and Phrases: CRISPR delivery, genome editing safety, off-target assessment, immunogenicity, clinical translation, risk modeling

1 INTRODUCTION

Introduction

CRISPR-based genome editing has moved from proof-of-concept to clinical reality, with approved and late-stage programmes targeting sickle cell disease, beta-thalassaemia, and hereditary transthyretin amyloidosis. This translation has been accompanied by a large and rapidly diversifying delivery literature: viral vectors, electroporation, biomaterials-mediated carriers, and ribonucleoprotein (RNP) delivery each promise distinct trade-offs between efficiency and safety. Several recent reviews of therapeutic delivery now routinely catalogue these modalities side by side, while parallel reviews document the persistence of off-target editing and immunogenicity as the two dominant safety liabilities of the field. What has not emerged is a way of putting these quantities on the same scale. Delivery efficiency is reported as editing percentage in a particular tissue at a particular dose; off-target risk is reported as a count of candidate sites or an AUPR score on a benchmark; immunogenicity is reported as seroprevalence in a cohort. The field therefore compares modalities and designs by argument rather than by measurement.

The core tension is that efficiency and safety are not independent axes but coupled ones, and the coupling runs through the delivery method itself. Viral vectors achieve durable expression precisely because the transgene persists, which is also why off-target events can accumulate over time; the same vectors provoke capsid-specific cytotoxic T-cell responses that clear transduced cells and erode editing persistence. Pre-existing immunity to Cas9 proteins is widespread but extraordinarily variable across studies, with anti-SpCas9 antibody prevalence reported from roughly 2.5% to 58% and anti-SaCas9 from 4.8% to 95%, and pre-immunisation has been shown to cause editing failure in vivo. Electroporation and RNP delivery

shorten exposure and reduce persistence-driven off-target risk, but introduce cytotoxicity and constrain dosing. Any metric that treats "efficiency" and "safety" as separately optimisable therefore misrepresents the design problem: the correct object is a joint quantity defined over a delivery-editing configuration.

A second obstacle is measurement heterogeneity. Off-target risk is assessed against benchmarks whose operational definitions of an "active" off-target site differ substantially: CHANGE-seq applies a cleavage-count threshold, whereas GUIDE-seq treats experimental detection as positive and generates negatives in silico, and both are dominated by negatives at ratios near 97%, so that accuracy appears near 0.99 while AUPR and F1 remain low. Prediction models built on these benchmarks inherit their label conventions and their class imbalance, and reported performance is consequently not comparable across studies. Immunogenicity measurement is similarly dispersed across assay types and populations. The consequence is that neither the benefit term nor the risk term of any putative objective function currently has a stable, transferable scale.

Quantitative risk frameworks developed outside biomedicine offer a route through this impasse. Probabilistic risk management converts qualitative risk registers into computable impact rankings by assigning distributions to each risk and propagating them by Monte Carlo simulation, yielding percentile-based absolute comparisons; the approach is explicit that results depend on expert elicitation and on an independence assumption that must be relaxed for coupled hazards. Engineering safety practice supplies intermediate variables that avoid premature probability estimation: ISO 13849-style risk graphs map hazards onto severity, frequency of exposure, and avoidability to produce a required performance level, and comparable severity-exposure-detectability decompositions underpin failure-mode scoring in high-significance systems. Clinical prediction research contributes the complementary machinery of calibration and net-benefit summarisation, in which a risk model is judged by the utility of the decisions it induces rather than by discrimination alone. Together these traditions suggest that a single scalar — benefit per unit risk — is constructible, provided the risk term is assembled from commensurable components and the benefit term is defined on the same configuration space.

Recent work in adjacent computational domains reinforces both the feasibility and the pitfalls. Safety cases for learned components argue that quantitative claims about model behaviour must be tied to explicit operational ranges and monitored at deployment [1]; evaluations of clinical and agentic AI show that aggregate benchmark scores are comparative rather than universal thresholds, and that aleatoric and epistemic uncertainty must be separated before any score is used to justify a decision. These lessons transfer directly: an off-target predictor's score is an epistemic estimate conditioned on a benchmark, not a property of the nuclease, and an immunogenicity prevalence is an aleatoric population parameter, not a fixed attribute of a vector. Treating them as interchangeable inputs to a single formula is the principal failure mode we seek to avoid.

This review differs from existing surveys in three respects. First, whereas delivery reviews organise evidence by modality and safety reviews organise evidence by hazard type, we organise it by the quantity that a unified objective must contain, and we show where the literature supplies a commensurable value and where it supplies only a qualitative claim. Second, we treat benchmark construction as part of the measurement problem rather than as background: the divergence between CHANGE-seq and GUIDE-seq label semantics, and the extreme negative imbalance of both, are shown to propagate into any risk term naively derived from predictor performance. Third, we import formal devices from quantitative risk and safety engineering — distributional risk registers, severity-exposure-avoidability triples, calibration-and-utility assessment, and uncertainty decomposition — and test each for transferability rather than asserting analogy.

Contributions

- We formalise the target quantity as editing benefit per unit risk over a joint delivery-editing configuration space, and show that the four dominant delivery modalities differ not in which safety hazards they incur but in the exposure-duration and vector-intrinsic parameters through which they incur the same two hazards, off-target editing and immunogenicity. - We assemble the causal chain from vector immunogenicity through immune clearance to loss of editing persistence, and demonstrate that its links are individually evidenced but have not been integrated into a single risk model, leaving immunogenicity as an unparameterised input in current design decisions. - We audit the off-target benchmark field and show that the two de facto standard benchmarks rest on incompatible operational definitions of an active site and share an approximately 97% negative class balance, so that accuracy, AUPR, and F1 imply systematically different risk scales and cannot be pooled without normalisation. - We map the limitation structure of the field onto a two-dimensional coordinate of lifecycle stage by hazard nature, finding that the majority of documented limitations fall in the validation-and-deployment stage under evidentiary uncertainty, which identifies where a quantitative objective would contribute most and where evidence

is thinnest. - We establish a graded transferability account of quantitative safety instruments, showing that uncertainty dichotomisation and resampling-based confidence intervals transfer to CRISPR off-target assessment almost without loss, that attack-success-style rates transfer only after semantic substitution and direction reversal, and that the principle that aggregate scores are comparative rather than universal thresholds transfers completely while exposing the deepest measurement gap in the field. - We specify how distributional risk registers, severity-exposure-avoidability triples, and calibration-and-utility assessment can be composed into a single computable objective, and identify the independence assumption in conventional risk-register arithmetic as the specific obstacle that a structured hazard graph must replace. - We derive design implications for reporting practice, arguing that delivery efficiency, off-target risk, and immunogenicity must be reported jointly with their uncertainty type and benchmark provenance so that benefit-per-risk comparisons across modalities become reproducible.

2 BACKGROUND AND PRELIMINARIES

This section fixes the vocabulary used throughout the review. Because the review spans three normally separate literatures—CRISPR delivery engineering, quantitative safety assessment, and uncertainty modeling for large language models (LLMs)—we first define the concepts that recur across the thematic chapters, then state how they relate. Each subsection gives a working definition, traces the concept's origin or evolution, and cites representative sources. The definitions are deliberately operational: they are phrased so that later chapters can attach measurements, benchmarks, and objective functions to them without redefinition.

For the above reporting requirements to be truly operational, it is first necessary to clarify the conceptual tools that run through CRISPR delivery engineering, quantitative safety assessment, and uncertainty modeling, and to specify their interrelationships.

2.1 Editing Efficiency and Editing Safety

Editing efficiency denotes the fraction of target cells (or alleles) that acquire the intended edit under a given delivery condition; editing safety denotes the complement—the probability and severity of unintended outcomes, including off-target editing, on-target but undesired repair outcomes, and immune or toxic responses. The two are conventionally reported as separate endpoints, but recent engineering surveys treat them as jointly optimizable properties of the same delivery-and-enzyme system. Clinical progress reports show that the therapeutic window is set not by either quantity alone but by their ratio at a specified dose. Off-target effects have accordingly been reframed from a nuisance to a measurable safety endpoint with its own evaluation methodology, and computational guide-design work now treats off-target safety as a first-class prediction target rather than a post hoc filter [2]. We therefore define a risk-adjusted editing yield as the intended-edit rate discounted by a normalized off-target risk score; this composite is the quantity the review's synthesis chapters attempt to compare across studies.

2.2 Delivery Modality and Dose

A delivery modality is the physical or biochemical route by which editing components enter cells: viral vectors (notably AAV), electroporation, lipid nanoparticles (LNPs), and pre-formed ribonucleoprotein (RNP) complexes. Each modality is characterized by a carrier capacity (cargo size limit), a dose (particles, volts, or molar amount per cell), and a persistence profile (duration of editor expression). Systematic reviews of in vivo delivery document that these attributes trade off against one another: AAV offers durable expression but limited capacity and immunogenicity, whereas RNP delivery is transient and less immunogenic but requires repeated dosing. Disease-specific reviews reach the same conclusion in oncology and CNS contexts, where blood–brain barrier crossing further constrains modality choice. We treat dose, capacity, and persistence as the three coordinates of a delivery configuration, so that a study's reported efficiency and safety can be conditioned on its configuration rather than compared naively.

2.3 Immunogenicity as a Persistence Attenuator

Immunogenicity is the propensity of a delivery vehicle or editing enzyme to elicit an adaptive or innate immune response. It has two distinguishable sources: vector immunogenicity (capsid or lipid components) and editor immunogenicity (bacterial-derived Cas proteins). Work on reducing nuclease immunogenicity shows that both sources shorten the window of effective editing and can produce adverse reactions independent of off-target activity. Enzyme size is a further coupling variable, since larger editors strain

carrier capacity and thus restrict modality choice. We model immunogenicity as a decay factor on editing persistence: the effective duration of editing is the nominal persistence multiplied by an immunogenicity-dependent attenuation term. This formulation lets immune responses enter the same objective function as efficiency and off-target risk, rather than being reported as a separate safety note.

2.4 Off-Target Benchmarks and Risk Scales

An off-target benchmark is a dataset plus a scoring procedure used to rank candidate guide RNAs or editing outcomes by their likelihood of unintended editing. Benchmarks differ in cell type, assay (e.g., GUIDE-seq-like versus computational prediction), and metric, so raw scores are not comparable across sources. The evaluation literature has made the general point that a benchmark's metric encodes a value judgment about what counts as safe, and that generic accuracy, recall, and precision do not by themselves express a clinically acceptable threshold. The same concern appears in clinical evaluation benchmarks, where safety and effectiveness must be scored on separate, explicitly defined axes [3]. We therefore distinguish a raw off-target score (benchmark-specific) from a normalized off-target risk score (mapped to a common scale, e.g., predicted probability of a clinically reportable off-target event). Only the latter is admissible in cross-study comparison.

2.5 Uncertainty, Risk, and Translation

Uncertainty in this review follows the standard split between aleatoric uncertainty (variability inherent to the assay or population) and epistemic uncertainty (ignorance about the model or mechanism). LLM safety research has developed operational instruments for this split—attack success rate, calibration measures, and risk-aware evaluation frameworks—that are defined independently of any particular model. Translation frameworks supply the second ingredient: a structured mapping from qualitative limitations and assumptions to quantitative risk statements, as proposed for AI-to-clinical translation. We adopt a three-part scheme: a limitation is an observed constraint on evidence (e.g., a benchmark validated only in one cell line); an assumption constraint is a condition under which a claim holds (e.g., chromatin accessibility at the target locus); and a risk element is a limitation or assumption mapped to a probability-and-severity pair. This scheme is what allows the review to convert the qualitative caveats scattered through the CRISPR literature into inputs for the unified benefit–risk objective introduced in the thematic chapters.

Taken together, these five concept clusters form a single chain: a delivery configuration sets dose, capacity, and persistence; immunogenicity attenuates persistence; off-target benchmarks supply the risk score that discounts yield; and the uncertainty-and-translation vocabulary specifies how each quantity is to be measured, compared, and reported. Later chapters instantiate this chain for specific modalities and evidence sources.

3 RELATED WORK

3.1 Delivery-Centric Views of CRISPR Therapeutics

The clinical success of CRISPR-based interventions depends primarily on the efficient delivery of editing components into target cells, and a substantial body of review literature has organized itself around this dependency. Early comparative surveys catalogued viral vectors, electroporation, and biomaterials-mediated delivery as parallel routes, each with distinct efficiency profiles. Subsequent work sharpened this comparison by contrasting viral and non-viral strategies for specific disease contexts, including cancer [4] and neurological disorders, while polymer-based and extracellular-vesicle platforms were advanced as safer alternatives to nuclease delivery [5, 6]. Biomaterials-mediated delivery has been reviewed as a distinct design space with its own challenges in infectious disease applications, and engineered exosomes have been proposed as precision immunotherapy vectors. Editorial commentaries have consolidated these threads, framing delivery as the rate-limiting step for translation. Across this literature, delivery efficiency is the organizing axis, and safety enters largely as an ancillary caveat. Our review departs from this framing: rather than treating delivery as a standalone engineering problem, we position vector choice as a risk-bearing decision whose consequences for off-target activity and immune response must be modeled jointly.

3.2 Mechanistic Accounts of Off-Target Activity

A second and larger stream examines off-target effects as a mechanistic phenomenon to be detected and described. Foundational reviews characterize off-target cleavage as an intrinsic property of programmable

nucleases and survey the mismatch tolerance that governs it. This line has been extended into comprehensive treatments of CRISPR mechanisms and their clinical challenges in oncology, and into disease-specific reviews for hypercholesterolemia [7] and Duchenne muscular dystrophy. A recurring theme is mitigation: reviews enumerate high-fidelity Cas variants, truncated guide RNAs, and delivery-timing strategies as countermeasures. Base editing has been presented as a partial structural remedy because it avoids double-strand breaks [8], and broader gene-editing technology surveys situate precision refinement within this trajectory. The explanatory power of this literature is considerable, yet its output is predominantly qualitative: it tells us which mechanisms generate off-target events, not how to convert detection signals into comparable risk magnitudes. Our work retains the mechanistic vocabulary but redirects it toward quantitative risk scoring.

3.3 Computational Prediction of Off-Target Effects

Within the off-target stream, a computationally oriented school treats prediction as a supervised learning problem. Early ensemble and feature-engineering approaches established that sequence and chromatin features carry predictive signal. Deep learning then reshaped the field: multi-view networks model guide–target interactions across heterogeneous representations, genome-wide predictors target improved generalization [9], and versatile tools package these models for practitioner use. More recently, pretrained language representations have been adapted to the guide-RNA setting [10], and explainability has become an explicit design goal. Parallel efforts predict on-target efficiency and off-target behavior for Cas13d [11]. This body of work demonstrates that off-target propensity is learnable, but its evaluation conventions—accuracy, recall, AUC—are benchmark-local and do not translate into clinically interpretable thresholds. We build on these predictors as feature sources while arguing that their outputs require calibration into a risk scale rather than direct reporting.

3.4 Detection Benchmarks and Their Comparability

A closely related empirical tradition concerns how off-target events are measured. Reviews of genome-wide detection techniques compare GUIDE-seq, CHANGE-seq, and related assays on sensitivity and coverage, and methodological surveys emphasize that assay choice materially changes the observed off-target field. The detection literature has matured to the point where benchmark datasets are reused across studies, yet cross-benchmark normalization remains largely unaddressed; different laboratories report incompatible summary statistics for nominally the same construct. This comparability problem is central to our argument: we treat heterogeneous benchmark outputs as inputs to a unified risk scale, an operation the detection literature itself does not attempt.

3.5 Immunogenicity as an Under-Integrated Dimension

Immunogenicity constitutes a third safety axis whose review coverage is markedly thinner. The most direct evidence comes from AAV biology, where capsid-specific cytotoxic T-cell responses clear transduced cells and produce dose-dependent loss of transgene expression. Clinical immunogenicity risk assessment for novel biotherapeutics has been reviewed as a preclinical-to-clinical translation problem [12], and design-based minimization strategies have been catalogued for antibody therapeutics. For cell therapies, immunogenicity heterogeneity across donors and manufacturing routes has been flagged as a translational obstacle [13], and gene-edited hypoimmunogenic organoid platforms have been profiled for their immune and mutational signatures. A dedicated treatment of CRISPR therapeutic immunogenicity argues that innate and adaptive responses to Cas proteins and delivery vehicles are critical considerations [14]. Notably, these strands rarely connect to the off-target literature; immunogenicity is discussed as a vector property, not as a modifier of editing outcomes. Our framework explicitly links vector immunogenicity to editing persistence and downstream risk, closing a gap this literature leaves open.

3.6 Clinical Translation Roadmaps and Their Limits

A translational school organizes the field around barriers to clinical adoption. The most prominent example combines delivery strengths and limitations into an explicit roadmap for clinical translation, and analogous roadmaps exist for blood disorders [15] and for AAV-based gene therapy more broadly. Systematic reviews have extended this lens to antimicrobial applications, and comparative analyses of delivery systems for rare metabolic disease illustrate the trade-off reasoning characteristic of the genre. Science-policy scholarship adds an innovation-system perspective on how funding shapes which translational paths are pursued. These

roadmaps are valuable as agendas but remain qualitative checklists; they identify what must be solved without specifying how competing risks should be weighed. Our contribution is to convert this checklist into a structured, quantifiable preclinical safety framework.

3.7 Risk Modeling Traditions Outside Biomedicine

Because the CRISPR literature lacks a quantitative risk-modeling tradition, we draw on adjacent fields where such traditions exist. In engineering safety, quantitative assessment methods have been developed for vehicle safety risk [16], scenario-based validation of automated driving [17], and residual-risk validation [18]. Formal treatments of quantitative safety and liveness provide the conceptual vocabulary for distinguishing acceptable from unacceptable behaviors. Reliability and assurance arguments for machine-learning components show how statistical evidence can be assembled into a safety case [19], and activation-range supervision offers a concrete mechanism for runtime safety monitoring [20]. Occupational and process-industry studies contribute dynamic risk modeling and quantitative risk analysis practices, while project-risk scholarship critiques probability–impact matrices in favor of quantitative prioritization [21]. Governance frameworks for pension and security risk illustrate institutionalization of risk scoring. These works supply the methodological scaffolding our framework adapts, though none addresses biological editing hazards.

3.8 Clinical Risk Prediction and Its Evaluation

Medical risk prediction offers a second methodological reservoir. Scoping reviews of surgical risk models trace the shift from traditional scoring to artificial-intelligence-based prediction and document the persistent challenge of clinical translation. Evaluation methodology in this space emphasizes clinical utility rather than discrimination alone, exemplified by the weighted Brier score [22]. Risk-aware frameworks have been proposed for clinical text de-identification and re-identification risk [23], and safety risk assessment frameworks exist for children's online safety. Quantitative frameworks for structural reliability integrate conformity assessment with partial safety factors. What this literature demonstrates is that risk scores become actionable only when paired with explicit utility considerations and calibrated thresholds; we adopt this lesson for CRISPR safety, where no analogous calibration currently exists.

3.9 LLM Safety Evaluation as a Parallel Enterprise

Large language model safety has developed its own review ecology, largely disconnected from the biomedical safety literature. Survey work has mapped evaluation-safety failures and proposed conceptual frameworks for diagnosing them. Benchmark construction has proceeded through diverse risk taxonomies and guardrail datasets [24], and domain-specific evaluations now target clinical utility and safety jointly. Frameworks for evaluating safety risks in real-world applications argue that foundation-model testing is insufficient and that deployment context must be modeled [1]. Child-safety protection frameworks extend this to vulnerable populations [25], and embodied-agent studies examine subtle risks in decision-making settings. Uncertainty quantification has emerged as a frontier concern, with the aleatoric–epistemic distinction and bootstrap-based interval estimation receiving explicit attention. Translation-quality estimation work shows how LLM outputs can be scored against calibrated criteria. This literature is methodologically rich but is not represented in our evidence set applied to biological editing risk; our review is, to our knowledge, the first to import its uncertainty vocabulary into CRISPR safety assessment.

3.10 LLM Assistance for Gene-Editing Workflows

A small but growing set of studies places language models directly in the gene-editing loop. Agentic automation of editing experiments has been demonstrated with LLM-driven planning and execution, signaling that models may soon mediate experimental design decisions that carry safety consequences. General-purpose deep learning frameworks for scientific and clinical data provide the infrastructure on which such agents depend [26], and domain-specific predictors of off-target binding for engineered receptors illustrate the broader trend toward learned safety surrogates. However, these efforts optimize task performance and do not incorporate calibrated uncertainty or risk thresholds. The intersection of LLM assistance with quantitative safety modeling for editing remains essentially unexplored, which is precisely where our framework intervenes.

3.11 Adjacent Quantitative Safety Practices

Several further strands inform our design without directly addressing genome editing. Crowd-safety modeling demonstrates how agent-based simulation can quantify emergent hazard in collective settings, and driver-training evaluation shows how intervention effectiveness can be measured against safety outcomes [27]. Integrated model-based machine safety in flexible production illustrates how safety and quality objectives can be co-optimized in a single framework, a logic we transfer to the efficiency–risk trade-off in editing. Marine route-decision safety assessment contributes geometric risk formulations [28], and highway driving risk assessment offers composite potential-field representations of hazard [29]. Risk-aware trajectory prediction under safety-critical scenarios further exemplifies probabilistic hazard forecasting. Common-cause failure analysis in safety-critical digital instrumentation provides a template for quantifying correlated failure modes, and terminology frameworks for assessing habitability illustrate how quantitative constructs are defined and bounded [30]. Together these works establish that quantitative safety assessment is a mature discipline elsewhere; their absence from CRISPR discourse is a structural gap rather than a conceptual impossibility.

3.12 Application-Domain Reviews and Their Safety Treatment

Disease-specific reviews form a final group that reveals how safety is typically handled at the application level. Reviews of CRISPR medicine for blood disorders discuss safety as one challenge among many, and cancer-focused surveys treat off-target effects as a secondary concern relative to efficacy. Bacterial-immunity reviews provide the evolutionary backdrop for editing systems, while drug-delivery surveys emphasize formulation over hazard. Neurological and muscular applications similarly foreground delivery and efficacy. The pattern is consistent: safety is acknowledged, rarely quantified, and almost never integrated with delivery and immunogenicity into a single assessment. This consistency across domains strengthens the case for a cross-cutting framework rather than another application-specific review.

3.13 Synthesis and the Present Contribution

Taken together, the literature divides into mature but siloed streams: delivery surveys optimized for efficiency, mechanistic off-target reviews oriented toward detection and mitigation, computational predictors evaluated on benchmark-local metrics, a thin immunogenicity literature disconnected from editing outcomes, and qualitative translation roadmaps [15]. Quantitative risk-modeling traditions exist in engineering and clinical prediction, and LLM safety evaluation has developed sophisticated uncertainty and benchmark practices [1], but the two have never met the CRISPR safety literature. Three blind spots follow: no framework places delivery efficiency and editing safety in a single objective; no framework normalizes heterogeneous off-target benchmarks into a comparable risk scale; and no framework connects LLM-assisted prediction with calibrated uncertainty for editing hazards. Our review addresses all three by proposing a quantifiable preclinical safety framework that couples delivery, off-target, and immunogenicity signals and by explicitly positioning LLM-assisted risk prediction—with aleatoric and epistemic uncertainty made first-class—as the integrative layer that existing reviews lack.

- [31] — delivery strengths, limitations, and a clinical translation roadmap.
- [32] — capsid-specific T-cell responses linking vector immunity to loss of transgene expression.
- [14] — innate and adaptive immune considerations for CRISPR therapeutics.
- [33] — genome-wide off-target detection and its clinical implications.
- [10] — pretrained sequence representations for off-target prediction.
- [9] — genome-wide prediction and guide optimization.
- [34] — LLM agents for automating gene-editing experiments.
- [1] — context-aware safety risk evaluation for real-world LLM applications.
- [22] — clinical-utility-weighted evaluation of risk prediction models.
- [17] — an integrated safety integrity framework from a safety-critical domain.

4 SURVEY METHODOLOGY

Of the 139 screened publications, 52 are cited in the synthesis; the remaining 87 informed screening and background and are not individually cited.

We assembled a corpus of 139 publications through multi-source retrieval, prioritizing arXiv and falling back to open academic indexes when arXiv coverage was insufficient. Deduplication was anchored on DOI and other persistent identifiers, so that each record entered the corpus exactly once and remained traceable to its canonical source. The resulting set defines the evidentiary boundary of this review: all subsequent mapping, screening, and evaluation operate strictly within these 139 papers, and no material outside this boundary is introduced at later stages.

Once the cross-domain conceptual tools and their interrelationships have been clarified, how to operationalize these tools into a verifiable evidence base becomes the precondition for further analysis; and the establishment of this evidence base depends, first of all, on how the literature collection is constructed and how its boundaries are delineated.

Refinement proceeded against a fixed protocol rather than an evolving one. The review is organized around 5 research questions decomposed into 15 sub-questions, and the evidence set for each sub-question was frozen before any analysis began. Screening therefore served to assign papers to sub-questions and to record their provenance, not to adjust the questions themselves. This ordering matters because it prevents the common failure mode in which the availability of evidence quietly reshapes the questions being asked; here the questions are stipulated first, and the corpus is then interrogated against them.

For each retained paper we extracted six evidence categories—problem, method, dataset or benchmark, metric, limitation, and assumption or constraint—together with the cross-work links that connect them. This produced a structured evidence matrix in which papers, claims, and methodological components are jointly represented. The six classes were chosen to cover both what a study addresses (problem, dataset, metric) and what it presupposes or fails to establish (limitation, assumption), since the latter frequently determine whether reported results transfer beyond their original setting.

Evaluation was conducted at three levels. First, evidence-matrix construction consolidated the extracted evidence categories and links into a single queryable structure. Second, each of the 15 sub-questions was analyzed against this matrix using the operation appropriate to it—classification, comparison, temporal evolution, or trade-off analysis—so that the form of the answer follows the form of the question rather than a uniform template. Third, every conclusion was decomposed into individually checkable claims, each annotated with the specific papers that support it, allowing any statement in the review to be traced back to its evidentiary basis and contested on that basis.

This three-level arrangement separates concerns that are often conflated: evidence-matrix construction is a representation step, sub-question analysis is an interpretive step, and claim-level verification is an accountability step. The separation allows a reader to accept the representation while disputing an interpretation, or to accept an interpretation while checking whether the cited papers actually support it. Because the evidence set was fixed in advance and each claim carries explicit paper-level attribution, the review's conclusions remain auditable against the original 139-paper corpus.

5 SYNERGISTIC OPTIMIZATION OF CRISPR DELIVERY EFFICIENCY AND EDITING SAFETY

Now that the evidence set has been fixed and can be traced back item by item, the next question is how this evidence is distributed and related under specific research topics; therefore, it is necessary to move on to organization and analysis at the thematic level.

Having established the methodological apparatus, this chapter turns to the central tension that governs the field: the pursuit of higher CRISPR delivery efficiency is routinely entangled with elevated off-target activity and immune activation, yet the evidence linking delivery choices to safety outcomes remains fragmented and largely qualitative. Across viral vectors, electroporation, biomaterials-mediated delivery, and ribonucleoprotein delivery, we show that methods are not bound to distinct safety problems but rather occupy different regimes of expression duration and dosing, and that dose–efficiency–toxicity trade-offs surface as discrete problem, limitation, and assumption nodes rather than as quantified relationships. No study in the current evidence base formalizes delivery efficiency and editing safety into a single comparable scalar, so any unified benefit–risk ratio must be treated as operational and provisional. This unresolved quantification gap, together with the persistent under-reporting of immunogenicity, sets up the reconstruction of neglected immune pathways in the following chapter.

5.1 Delivery-Method Safety Signatures and the Expression-Duration Axis

Which safety liabilities do the principal delivery modalities—viral vectors, electroporation, biomaterial carriers, and ribonucleoprotein (RNP) complexes—carry in the published record, and how strong and how direct is the evidence linking each modality to those liabilities? Across the literature reviewed here, the answer is not a simple ranking but a structured pattern: viral vectors exhibit the most explicit and best-documented safety associations, biomaterials occupy an intermediate position in which safety is framed as mitigation rather than elimination, electroporation's safety profile is largely implicit and must be inferred from mechanistic accounts, and RNP delivery consistently emerges as the modality with the weakest association to genotoxic and immunogenic harm.

Viral vectors show the most direct and consistently documented safety associations of the four modalities, with immunogenicity and insertional mutagenesis standing out as the two liabilities most frequently tied to them. Reviews of gene-editing delivery converge on the view that viral carriers combine high transduction efficiency with an intrinsic immunogenic burden and a risk of genomic disruption at the integration site [35]. The strength of this association reflects both the maturity of the vector platforms and the volume of clinical experience accumulated with them; adeno-associated virus (AAV) in particular has been characterized extensively enough that its immunological limitations are now treated as a defining property rather than an incidental finding [32]. This contrasts with the other modalities, whose safety liabilities are typically described in general mechanistic terms rather than documented as recurrent clinical observations.

The immunogenicity of viral delivery is not a marginal concern but a prevalence-level constraint on patient eligibility. Between 30% and 50% of the human population carries pre-existing neutralizing antibodies against AAV, a fact that has contributed to several companies scaling back or abandoning AAV-based gene therapy programs [31]. The picture is equally unfavorable for the editing payload itself: seroprevalence of anti-SpCas9 antibodies ranges from 2.5% to 58% across studies, and anti-SaCas9 prevalence from 4.8% to 95% [14]. These figures indicate that the immune barrier operates at two levels—against the vector and against the nuclease—and that the two compound each other in viral systems, where both components are presented to the host simultaneously. Biomaterial and RNP approaches, by contrast, can at least in principle decouple these two sources of immunogenicity.

A second, mechanistically distinct liability of viral vectors is the persistence of expression and its consequences for off-target editing. Because AAV-delivered genes can remain expressed in terminally differentiated cells for years, the editing machinery stays active long after the intended modification, creating opportunities for off-target mutations to accumulate over time [36, 35]. This temporal dimension is easily overlooked in comparisons that report only instantaneous editing efficiency, yet it reframes the safety question: the relevant quantity is not the per-unit-time off-target rate but the cumulative exposure of the genome to active nuclease. The same logic applies to plasmid-based delivery, which likewise sustains expression, and it is this shared property—not any feature of the vector chemistry—that unites the two platforms in the off-target literature.

Electroporation presents a different evidentiary pattern: its documented limitations concern delivery performance rather than genotoxicity, and its safety profile must be reconstructed from physical mechanism rather than from direct clinical reporting. Comparative assessments of CRISPR-Cas9 delivery systems describe electroporation primarily in terms of efficiency, applicability, and the practical constraints of ex vivo manipulation, with cytotoxicity and physical damage to cells mentioned as consequences of the electric field rather than as characterized safety endpoints [35]. The association with safety is therefore real but indirect: it follows from the biophysics of membrane permeabilization rather than from a body of reported adverse events. This distinguishes electroporation sharply from viral delivery, where the safety association is explicit and repeatedly observed, and it suggests that the field's evidence base is unevenly distributed across modalities.

Biomaterial-mediated delivery occupies an intermediate position, in which safety is explicitly addressed but framed as attenuation rather than elimination. Polymeric and lipid carriers are consistently presented as responses to the immunogenicity and cytotoxicity of viral systems, and their design objectives are stated in those terms. Yet the same literature acknowledges that lipid nanoparticles can still induce dose-dependent hepatotoxicity and pro-inflammatory cytokine release, so the safety gain is relative rather than absolute [31, 37]. The distinction matters for how the evidence should be read: biomaterials shift the safety burden from immunogenicity against a foreign capsid to tolerability of the carrier itself, a trade that is favorable but not free of cost.

RNP delivery stands out as the modality with the weakest association to safety liabilities, and the quantitative evidence on this point is comparatively strong. In an aggregated analysis of 67 studies, RNP

delivery was associated with the lowest off-target rates, typically below 1%, against 5–15% for plasmid-based delivery, while also showing lower immunogenicity and cytotoxicity [31, 36]. The magnitude of this difference—roughly an order of magnitude in off-target frequency—is large enough to reframe delivery choice as a safety decision rather than merely a logistical one. It also explains why RNP and mRNA formats have become the preferred vehicles in clinical applications where editing precision is paramount.

These disparate observations resolve into a single organizing principle: the direction in which a delivery method acts on safety is governed by the duration of nuclease expression. Persistent formats—plasmids and viral vectors—sustain activity and therefore carry higher off-target risk, whereas transient formats—mRNA and pre-formed RNP complexes—limit the window of exposure and correspondingly reduce that risk [36, 35, 31]. This axis cuts across the conventional viral/non-viral dichotomy and explains anomalies that the dichotomy cannot: it accounts for why plasmid delivery resembles viral delivery in off-target terms despite lacking a capsid, and why biomaterial carriers can be either persistent or transient depending on what they encapsulate. Expression duration, rather than vector class, is the parameter that predicts the safety signature.

The transient-expression advantage is not fixed, moreover, but can be amplified through engineering of the editing agent itself. Chimeric cytosine base editors illustrate the point: the AID-3S variant, incorporating a tCDA1EQ fusion, reduced R-loop-associated off-target editing by approximately 13.7-fold relative to the original Target-AID construct, while AID-2S achieved an approximately 4.5-fold reduction, in both cases without sacrificing on-target efficiency. That such gains are attainable within an already transient format indicates that delivery kinetics and editor architecture are complementary levers rather than substitutes. The practical implication is that the safety ceiling of a delivery method is set jointly by how long the cargo acts and by how processively the cargo behaves while active.

Two boundaries qualify this account. First, the strength of the evidence is uneven: viral vectors are supported by prevalence data and clinical experience, RNP by aggregated comparative statistics, biomaterials by mechanistic argument, and electroporation largely by inference from physical principles [37, 38]. Comparisons across modalities therefore rest on heterogeneous evidence types, and the apparent clarity of the ranking partly reflects how thoroughly each modality has been studied rather than how safe it is in absolute terms. Second, the transient formats are not free of immune recognition, since the nuclease protein and its delivery vehicle can still provoke responses, and the seroprevalence data cited above apply to the Cas proteins irrespective of how they are introduced [14].

Taken together, the evidence supports a graded rather than categorical picture of delivery-associated safety. Viral vectors carry the most explicit and best-quantified liabilities, in the form of pre-existing immunity and persistent expression; biomaterials mitigate but do not remove immunogenic and cytotoxic burden; electroporation's risks are mechanistically plausible yet sparsely documented; and RNP delivery shows the most favorable profile, with off-target rates an order of magnitude below plasmid formats. The unifying variable is expression duration, which predicts safety outcomes better than the viral/non-viral distinction and which can be further modulated by editor engineering. This raises the question pursued in the following section: whether these delivery-level safety gains can be made to reinforce, rather than trade against, editing efficiency when the two objectives are optimized jointly.

5.2 Dose–Cargo–Safety Coupling and Its Conditional Trade-off Structure

Whether delivery dose and vector cargo capacity impose extractable trade-offs against editing efficiency, off-target burden, and immunogenicity is the question this section addresses. The evidence assembled here indicates that such trade-offs are real but conditional: they are neither universal across platforms nor uniformly represented in the literature, and their direction depends on whether the delivery vehicle sustains expression, on the molecular size of the editor, and on the pre-existing immune status of the host. We therefore treat the trade-off not as a single curve but as a family of platform-specific couplings, and we examine where the literature states them explicitly and where it leaves them implicit.

The clearest and most consistently documented coupling links persistent expression to simultaneous gains in efficiency and cumulative off-target risk. DNA-based vectors sustain nuclease expression over extended periods, which raises the probability of editing at the intended locus but equally extends the window during which low-affinity genomic sites can be engaged [35]. Off-target accumulation under such conditions is not a fixed property of the nuclease but a time-integrated consequence of continued activity, which is why prolonged expression and elevated off-target burden co-vary rather than compete [36]. Mitigation strategies that shorten the activity window therefore improve safety without demanding a reduction in delivered dose, a point that reframes the trade-off as one of expression duration rather than of quantity alone. This distinction matters because it implies that dose reduction is not the only lever available when

off-target risk must be controlled.

A second coupling operates through immunogenicity and is markedly steeper than the efficiency relationship. In a systematic assessment of recombinant AAV vectors, capsid-specific immune responses scaled monotonically with dose: high-dose AAV8-hFIX administration produced elevated liver enzymes together with declining transgene expression, whereas low doses elicited only mild, steroid-responsive inflammation [32]. The mechanism is self-reinforcing, since immune-mediated clearance of transduced cells removes the very template on which efficacy depends, so that dose escalation beyond a threshold degrades both safety and efficiency simultaneously. This is qualitatively different from the expression-duration trade-off described above, where the two objectives move in the same direction; here they collapse together, producing a non-monotonic efficacy curve that no simple dose optimization can escape.

The dose magnitudes required by systemic indications make this collapse a practical rather than theoretical concern. For Duchenne muscular dystrophy, body-wide treatment demands AAV doses in the range of $5.5 \times 10\hat{}14$ to $1.8 \times 10\hat{}15$ vg/kg, yet AAV9 at or above $1.5 \times 10\hat{}14$ vg/kg has already been associated with hepatic toxicity and renal injury in animal models, and clinical dosing between $5 \times 10\hat{}13$ and $3 \times 10\hat{}14$ vg/kg has been accompanied by cardiopulmonary insufficiency, thrombocytopenia, and death. The therapeutic window for systemic muscle editing is thus narrow and, critically, lies below the dose required for uniform transduction. In contrast to local or ex vivo routes, where delivered dose and exposed tissue volume can be decoupled, systemic administration binds efficacy and toxicity to the same variable, which is why the trade-off appears most acute precisely in the indications with the largest unmet need.

Layered onto dose is an immunological admission threshold that operates independently of the amount administered. Pre-existing neutralizing antibodies against AAV capsids are present in roughly 30–50% of humans, with seroprevalence across broader panels ranging from 2% to 60%, and their presence blocks transduction irrespective of how much vector is given [31]. Capsid-directed humoral and cellular responses further restrict repeat administration, foreclosing the dose-fractionation strategies that might otherwise widen the therapeutic window [32]. The consequence is that a substantial fraction of patients is excluded before dose selection even becomes relevant, and for those admitted, the usable dose range is compressed by the prospect of boosting on re-exposure. Immunogenicity therefore functions less as a point on the dose–response curve than as a gate that determines whether the curve is accessible at all.

Non-viral lipid nanoparticle systems relocate the trade-off rather than eliminating it. Lipid nanoparticle components can induce dose-dependent hepatotoxicity and pro-inflammatory cytokine release, yet the transient nature of mRNA or ribonucleoprotein delivery substantially reduces cumulative off-target editing relative to persistent expression [31]. The same transience that limits off-target accumulation, however, constrains the achievable efficiency per administration, since editing must be completed within a brief activity window [36]. Comparative analyses of delivery modalities consistently place lipid nanoparticles on the opposite side of the trade-off from viral vectors: lower off-target burden and reduced immunogenicity against the vehicle, but greater sensitivity of efficiency to dose timing and formulation [35]. The trade-off is thus preserved in structure while its dominant axis shifts from dose-to-off-target toward dose-to-transient-toxicity.

Cargo format introduces a third coupling that is often conflated with vector choice. Ribonucleoprotein delivery typically yields off-target rates below 1%, plasmid delivery reaches 5–15% at highly homologous sites, and mRNA occupies an intermediate position while suffering from intrinsic instability [31]. Because these formats differ in both persistence and packaging burden, comparisons that hold vector constant but vary cargo already reveal a monotonic relationship between expression duration and off-target frequency. This suggests that a meaningful portion of what is attributed to vector class is in fact attributable to the molecular form of the delivered editor, an attribution problem that complicates any attempt to read a single trade-off curve off the literature.

Editor size imposes a further constraint that couples cargo capacity to dose and safety in a second triangle. Cytosine base editors measure approximately 5.2–6.3 kb and therefore exceed the roughly 4.7 kb packaging limit of AAV, forcing either split-vector architectures or the substitution of compact orthologues such as SaCas9 and CjCas9 [8]. Split constructs frequently require higher viral doses to achieve comparable transduction and often sacrifice efficiency in the process, so that the size constraint propagates directly into the dose and immunogenicity axes discussed above. Clinical experience with base editing reinforces the point, as the transition from early-generation to refined constructs has been accompanied by explicit attention to packaging and dosing compromises. Cargo capacity is therefore not a neutral specification but an upstream determinant of where a platform sits within the dose–safety space.

Editor engineering supplies a partial counterexample in which the efficiency–immunogenicity relationship is deliberately inverted. Efforts to reduce nuclease immunogenicity have repeatedly found that engi-

neered variants sacrificing catalytic or binding efficiency for immune evasion achieve their primary goal at the cost of editing performance [39]. This is the one coupling in the assembled literature that is stated as an explicit trade-off rather than inferred from co-varying measurements, and it is expressed qualitatively as a recognized design conflict rather than as a parameterized relationship. The contrast with the dose-driven couplings is instructive: where dose effects are continuous and quantifiable, the immunogenicity–efficiency conflict is currently documented only as a directional design constraint, which limits its incorporation into quantitative optimization.

Taken together, the literature supports three distinguishable couplings rather than one. Persistent expression trades efficiency against cumulative off-target burden; dose escalation trades efficacy against capsid immunity and organ toxicity, with collapse rather than exchange at high doses; and cargo size trades packaging feasibility against dose and efficiency through split-vector architectures. These couplings are conditional on route, platform, and host immune status, and they interact, since a size-driven dose increase feeds directly into the immunogenicity axis. The representation of these relationships in the assembled evidence is uneven: dose–toxicity and dose–immunogenicity relationships are reported with quantitative resolution, whereas the efficiency–immunogenicity conflict is recorded only as a qualitative design tension. This asymmetry indicates that the field has measured what is easily measured and left the editor-intrinsic trade-offs comparatively unparameterized, a gap that any joint optimization framework will need to address.

5.3 Constructing a Commensurable Objective for Delivery–Editing Combinations

Whether delivery efficiency and editing safety can be folded into a single objective function—a benefit–risk ratio or a risk-adjusted editing yield—so that heterogeneous delivery–editor combinations become comparable on one scale remains an open design question rather than a settled result. The corpus contains no explicit formalization that assigns delivery efficiency and editing safety to a common scalar; what exists instead are partial aggregations, qualitative decision matrices, and cross-platform comparison tables that each capture a slice of the problem. This section argues that a unified objective is constructible in principle but currently underdetermined in practice, because the benefit term is measured on incompatible scales and the risk term resists commensuration across off-target, immunogenic, and manufacturing axes. The evidence supports a layered rather than a monolithic formulation: a scalar is defensible within a narrow editor family, whereas across delivery platforms only a structured, multi-attribute comparison is presently warranted.

The closest existing approximation to a scalar objective comes from the specificity-scoring family, which aggregates on-target cleavability and genome-wide off-target probability into a bounded quantity. The CRISOT-Spec formulation, defined as $10/(10 + P\ off)$ with $P\ off = \sum_kN_kP_k$, compresses predicted off-target burden across the genome into a value on $(0, 1]$, thereby rendering editors rankable on a single axis. This is a genuine formalization, but it is a safety-only objective: delivery efficiency enters nowhere in the expression, so the score cannot distinguish a highly specific editor that transfects poorly from one that edits efficiently. The mechanism is instructive—the denominator accumulates per-site off-target probabilities weighted by mismatch counts—yet the absence of a benefit term means the formulation answers a narrower question than a benefit–risk ratio requires. A unified objective would need to multiply or otherwise couple this specificity term with a delivery-yield term, and no study in the corpus performs that coupling explicitly.

A two-dimensional on/off-target profile offers a more promising commensuration strategy because it preserves both axes rather than collapsing them prematurely. By plotting mean on-target editing rate against mean off-target editing rate, one comparison places distinct cytosine base editors on the same coordinate system and shows that AID-2S and AID-3S reduce R-loop off-target events by roughly 4.5-fold and 13.7-fold relative to Target-AID while maintaining comparable on-target efficiency [8]. The value of this representation is that it makes the trade-off visible without asserting a fixed exchange rate between the two axes; a risk-adjusted yield could be read off as a family of iso-benefit curves drawn over this plane. In contrast to the scalar specificity score, the profile accommodates the possibility that the optimal editor depends on the delivery context, since a low-efficiency/high-fidelity editor and a high-efficiency/moderate-fidelity editor occupy different regions of the same plane. What the profile does not supply is the weighting that would collapse it to a single number, and that weighting is precisely where delivery and safety must be jointly parameterized.

The benefit term itself proves resistant to standardization, which undermines any attempt to place delivery efficiency on a common scale. Reported efficiencies vary not only across platforms but across cell models for the same platform: a polymer system (HPAE-EB) achieves 55–67.3% GFP positivity

in HEK293 cells yet yields only 8.2% exon 80 excision with plasmid and 43.2% with RNP in RDEB keratinocytes, while maintaining viability above 85% throughout. This divergence is not measurement noise but a genuine context dependence—uptake, nuclear entry, and repair-pathway activity differ by cell type—and it implies that a benefit term must be indexed to a specified target cell and cargo format. Consequently, a risk-adjusted yield computed in one model system cannot be transported to another without re-estimating the benefit denominator. The same caution applies to the numerator of any ratio: an editing percentage obtained with plasmid DNA is not interchangeable with one obtained with preformed RNP, as the keratinocyte data make explicit.

Where a full scalar is unavailable, structured multi-attribute frameworks have emerged as the practical substitute, and their design reveals what a unified objective would need to contain. A three-part decision matrix organized around target-organ accessibility, cargo size, and acceptable safety risk ranks platforms by efficiency—lipid nanoparticles reaching 60–80% in liver and AAV achieving 30–80% in CNS and retina—and functions as the nearest qualitative analogue to a comparison framework [31]. A parallel cross-platform matrix compares AAV, AdV, LV, LNP, VLP, EV, and polymer systems across cargo capacity, expression duration, tissue targeting, immunogenicity, redosing capability, and CMC burden, concluding that no single platform satisfies all therapeutic requirements. Both frameworks are explicitly multi-criteria rather than scalar, and their convergence on the same conclusion—that trade-offs are irreducible—suggests that the field's implicit objective is vector-valued. A unified scalar could in principle be obtained by assigning weights to these attributes, but the weights would encode clinical priorities rather than biological measurements, and no study in the corpus justifies a particular weighting.

The risk axis presents a deeper obstacle to commensuration, because off-target, immunogenic, and manufacturing risks are not measured on compatible scales and in some cases are not standardized at all. Off-target datasets generated under different assays, thresholds, and reference genomes have been characterized as not meaningfully comparable, which means that cross-study safety comparisons lack a common reference standard. Immunogenicity compounds the problem: 2–60% of human sera contain antibodies that restrict AAV application, and neutralizing and total antibody assays are difficult to standardize [32], while pre-existing B- and T-cell immunity to SpCas9 and SaCas9 is reported across studies at prevalences ranging from 5% to 95% in the absence of a CLIA-grade assay [14]. These ranges are too wide to be treated as point estimates in a ratio; they are better understood as evidence that the risk term carries irreducible uncertainty. A risk-adjusted yield would therefore need to propagate interval-valued rather than scalar risk inputs, a requirement no current formulation satisfies.

The three constraint families that a risk-adjusted yield must respect—insufficient benefit, elevated off-target burden, and elevated immunogenicity—are not independent failure modes but instances of the same objective violated along different axes. Low transfection efficiency, off-target editing, and vector or nuclease immunogenicity each correspond to a distinct term falling below or above its admissible bound [5, 31]. Recognizing this unity clarifies why single-axis optimizations tend to displace rather than resolve problems: improving delivery by increasing dose can raise immunogenicity, and improving fidelity by constraining the editor can depress yield. The mechanistic implication is that the objective should be formulated as a constrained optimization rather than an unconstrained ratio, with explicit feasibility bounds on each axis. Such a formulation would not require a single exchange rate among risks, only that each constraint be evaluable—which returns the discussion to the standardization gaps identified above.

A defensible synthesis is therefore layered: scalar commensuration is achievable within a bounded editor family, whereas cross-platform comparison presently requires multi-attribute representation. Within base editing, the on/off-target plane and the specificity score together supply enough structure to rank variants and to define iso-benefit curves [8, 9]. Across delivery platforms, the decision matrix and the comparison table supply the attribute set but not the weights [31, 40]. The boundary between these regimes is set by whether the benefit and risk terms share a measurement protocol; where they do not, collapsing to a scalar would manufacture precision that the underlying data cannot support. This layered view also explains the asymmetry noted earlier: safety terms with established assays can enter a ratio, while editor-intrinsic trade-offs recorded only qualitatively cannot.

Two boundary conditions qualify this synthesis and mark where a unified objective would fail if applied naively. First, the cell-model dependence of delivery efficiency means that a ratio calibrated in HEK293 may invert in a primary or disease-relevant cell type, as the keratinocyte data demonstrate [5]; the objective must therefore be conditioned on target tissue rather than treated as a platform constant. Second, the immunogenicity ranges cited above are wide enough that a ratio computed from their midpoints could reverse under plausible assay choices [32, 14]. These are not arguments against formalization but arguments for reporting the objective with its conditioning variables and uncertainty bounds attached. A ratio presented without them would be less informative than the multi-attribute tables it replaces.

In sum, a unified objective function for delivery–editing combinations is constructible but not yet constructible well: the specificity score supplies a formal safety term, the on/off-target plane supplies a two-axis commensuration, and the decision matrices supply the attribute set, yet no study couples these into a single benefit–risk expression with standardized inputs. The gap is measurement infrastructure rather than mathematical imagination—off-target comparability, immunogenicity assay standardization, and context-indexed efficiency reporting are the missing primitives. Addressing them would allow the qualitative design tensions recorded in the literature to be converted into parameterized constraints, which is the precondition for any joint optimization of delivery efficiency and editing safety. The following section takes up the question of how such constraints behave when optimization is attempted jointly rather than sequentially.

6 THE ROOTS OF NEGLECTED IMMUNOGENICITY AND PATHWAYS TO RECONSTRUCTION

Once the constraints are parameterized, their actual performance in joint optimization depends on how the different efficiency and safety metrics constrain one another; therefore, it is necessary to shift from the optimization structure itself to a concrete examination of the trade-off relationships among these metrics.

Having established how delivery efficiency and editing safety can be jointly optimized, the preceding discussion leaves unresolved a more insidious failure mode: immunogenicity that erodes editing persistence without overt toxicity. This chapter argues that the causal chain linking vector immunogenicity to immune response, loss of edited-cell persistence, and clinical risk is supported by substantial evidence, yet remains fragmented across AAV immunology, nuclease protein engineering, and CRISPR delivery literatures, and has therefore never been assembled into a unified risk model. Extracting comparative evidence across AAV, LNP, electroporation, and RNP systems confirms that immunogenicity differs sharply by vector, while formalizing it as a multiplicative decay factor on persistence proves ontologically coherent but currently admits only semi-quantitative operationalization. These findings expose a deeper obstacle: the evidence needed to compare and integrate such heterogeneous sources is itself unevenly structured, a problem that the next chapter takes up directly.

6.1 Immunogenicity as an Unintegrated Causal Layer in Editing Durability Assessment

Does the existing literature establish integrable causal evidence linking vector immunogenicity, editor-enzyme immunogenicity, enzyme size, and vector safety to editing outcomes, and if not, why have these strands resisted consolidation into a unified framework for editing durability and clinical risk? The evidence assembled here indicates that each individual link is documented, yet the links are rarely measured on commensurable scales, in comparable temporal windows, or within the same study populations, so that a causal chain can be narrated but not parameterized.

The most direct causal evidence runs from vector immunity to loss of transgene expression. Capsid-specific cytotoxic T cell responses eliminate transduced cells and thereby erase the therapeutic or editing signal, and this relationship is dose-dependent: high-dose AAV8-hFIX administration produced elevations in liver enzymes accompanied by declines in circulating hFIX [32]. This is a genuine cause-and-effect observation rather than a correlation, because the temporal ordering of immune activation, hepatocellular injury, and expression loss is internally consistent within the same subjects. However, the same evidence base exposes a measurement problem: immune monitoring lacks standardized protocols and relevant samples are difficult to obtain, so estimates vary across studies in ways that cannot be attributed to biology alone [32]. The causal link is therefore established in direction but not in magnitude, which is precisely the form of evidence least amenable to incorporation into a quantitative risk model.

A second strand concerns pre-existing immunity to the editor itself, and here the literature reports a striking and consequential dispersion. Anti-SpCas9 antibodies have been detected in anywhere from 2.5% to 95% of healthy individuals, and T cell responses from 57% to 95% [14]. Such a range is too wide to be treated as a point estimate; it implies that immunogenicity must enter any risk framework as a distribution whose parameters depend on assay format, cohort geography, and prior exposure history. The functional consequence of this pre-existing immunity is not merely serological. Mice pre-immunized against SaCas9 showed hepatocyte apoptosis and failure of genomic editing following AAV-CRISPR transduction of the liver [14], establishing that pre-existing adaptive immunity can abolish editing in vivo rather than merely reduce its efficiency. In contrast to the vector-immunity literature, which measures loss of expression, this work measures loss of the editing event itself, and the two endpoints are seldom collected together.

The efficiency–immunogenicity trade-off constitutes a third link, and it is the one most often framed as a design problem rather than a risk problem. Low-immunogenicity Cas9 variants exhibit a 30–40% reduction

in transient expression levels, whereas HypaCas9 retains more than 95% editing efficiency while lowering antibody titers by 60% and T cell cytokine secretion by 75% [39]. The HypaCas9 result is important because it demonstrates that the trade-off is not fixed: immunoevasion and potency can be partially decoupled. Yet the same source identifies a systemic mismatch in how such variants are evaluated, since current assessment prioritizes short-term immunosuppression over long-term immune tolerance and lacks a standardized system for measuring tolerance [39]. An intervention that reduces acute antibody titers by 60% may therefore tell us little about whether editing persists over years, and durability is the outcome that matters clinically.

Enzyme size couples to immune potential through a physical constraint rather than an immunological one, and this coupling is frequently overlooked. The AAV packaging ceiling of roughly 4–5 kb means that conventional base editors exceed delivery capacity, forcing architectural compromises with immunological consequences. Truncating tCDA1EQ from 208 to 121 amino acids and pairing it with SaCas9 yielded a 4036 bp single-AAV system that simultaneously reduced R-loop-mediated off-target editing by 4.5- to 13.7-fold [8]. This is a rare case in which size reduction, delivery feasibility, and fidelity improve together, which suggests that the size–immunogenicity axis is partly a design variable. The immunological implication, however, remains inferential: smaller constructs permit lower vector doses, and lower doses in turn reduce capsid antigen load, but the reviewed studies do not measure this cascade end to end.

Route of administration introduces a fourth axis of variation and provides the sharpest contrast in the available evidence. Ex vivo delivery produced zero anti-Cas9 antibodies across 31 patients treated with exa-cel, whereas an in vivo sickle cell disease phase I trial detected anti-Cas9 antibodies in 70% of patients, with seropositivity associated with weakened editing durability [39, 14]. The mechanistic explanation is straightforward: ex vivo editing removes the editor before reinfusion and avoids exposing the recipient to capsid antigen, while in vivo delivery confronts both pre-existing and induced immunity simultaneously. This contrast is analytically valuable because it isolates administration route as a causal variable while holding the editing payload roughly constant, and it indicates that durability estimates derived from ex vivo programs cannot be transported to in vivo programs.

Set against these four strands, the reason consolidation has not occurred becomes clearer. The endpoints are incommensurable: transgene expression, editing frequency, antibody titer, cytokine secretion, and clinical durability are measured in different units, on different timescales, and often in different species. The temporal mismatch is especially damaging, since acute immunogenicity is sampled within weeks while editing persistence is a question of years, and no standardized tolerance-assessment system bridges the two [39]. Assay heterogeneity compounds the problem, because the 2.5–95% seroprevalence range reflects methodological variation as much as population difference [14]. A unified framework requires commensurable inputs, and the field currently supplies distributions whose spread is partly artifactual.

A further obstacle is that the most mechanistically informative studies are the least clinically representative. The demonstration that pre-immunized mice fail to edit after AAV-CRISPR delivery is causally decisive but rests on an immunization protocol that does not map onto natural human exposure histories [14]. Similarly, the HypaCas9 trade-off data establish that decoupling is possible without establishing that the decoupling persists in patients [39]. Meanwhile, the dose-dependent capsid immunity observed at high AAV8-hFIX doses is a strong clinical signal, but it concerns a transgene product rather than an editing enzyme, so its transfer to editing durability requires an assumption that the underlying cytotoxic mechanism is identical [32]. Each of these gaps is individually bridgeable; collectively they mean that no single study measures vector immunity, editor immunity, construct size, and long-term persistence in the same cohort.

The base-editing literature illustrates both the promise and the limits of co-optimization. The 4036 bp system shows that size, delivery, and fidelity can be addressed in a single design cycle [8], and this suggests a template in which immunological parameters are treated as design constraints from the outset rather than as post hoc safety findings. However, the same study reports off-target reduction without reporting immune readouts, so the immunological benefit of the smaller construct remains an inference from dose rather than a measured outcome. The pattern recurs across the corpus: studies that optimize one axis rarely report the variables needed to connect it to the others.

Taken together, the evidence supports a directional causal chain from vector and editor immunogenicity through editing failure to loss of durability, but not a quantitative one. The links are documented in isolation, measured on incompatible scales, and sampled over incompatible time horizons, and the assays that would permit pooling are not standardized [32, 14]. What is missing is not a hypothesis but a measurement architecture: a common set of immune readouts collected longitudinally alongside editing persistence, in both ex vivo and in vivo settings, with assay standardization sufficient to make seroprevalence estimates comparable. Until such a architecture exists, immunogenicity will continue to be reported as a safety

observation rather than integrated as a causal input to durability prediction. The next section turns from this diagnostic account to the constructive question of what such an assessment framework would need to contain.

6.2 Carrier-Dependent Immunogenicity: From Modality Differences to Comparable Control Evidence

Do the different delivery modalities—AAV, lipid nanoparticles, electroporation, and ribonucleoprotein complexes—map onto distinct immune-response types and intensities in a way that the literature supports as extractable, comparable evidence? The short answer is that they do, but only partially and asymmetrically: the corpus establishes a clear qualitative contrast between viral and non-viral delivery, while quantitative cross-modality comparison remains sparse and is usually inferred from safety annotations rather than measured head-to-head.

The strongest and most direct evidence chain concerns AAV. Recombinant AAV immunogenicity is treated in the literature as a well-defined, addressable problem rather than an incidental observation, and reviews devoted to the topic establish it as a first-order constraint on gene transfer. The mechanistic picture is unusually consistent across sources: pre-existing humoral immunity dominates. Between 2% and 60% of human sera contain anti-AAV neutralizing antibodies depending on capsid variant, and these antibodies are highly cross-reactive, so that seropositivity against one serotype frequently predicts neutralization of others. This is why pre-existing humoral immunity is repeatedly described as one of the most effective barriers to AAV delivery—not because titers are uniformly high, but because cross-reactivity collapses the escape space that capsid engineering might otherwise exploit.

A second, mechanistically distinct AAV constraint is adaptive memory. AAV transduction induces adaptive responses that establish immunological memory, and this memory directly obstructs readministration. The nonhuman primate evidence is particularly instructive because it isolates the effect: re-dosing with AAV8 failed to increase transgene expression while anti-AAV antibody titers rose. The failure is therefore not one of vector potency but of host priming—the second dose is neutralized before it can act. This distinguishes AAV from modalities whose limitation is transient and dose-local, and it explains why AAV protocols so often resort to immunosuppression rather than reformulation.

Lipid nanoparticles present an almost mirror-image profile. Their principal immune signature is innate stimulation—RNA and lipid components engaging TLR and interferon pathways—rather than adaptive priming. Reviews of nuclease immunogenicity accordingly frame LNP-mRNA as reducing adaptive immune activation while still requiring management of innate stimulation. The practical consequence is significant: unlike viral vectors, repeated in vivo dosing with LNPs is not reported as problematic with respect to induction of adaptive immunity. Two independent demonstrations support this. LNP delivery achieved 70% downregulation of Serpinc1 with no immune issues on repeat dosing, and low-immunogenicity LNPs supported repeated CRISPR-Cas9 mRNA administration to skeletal muscle. The contrast with AAV re-dosing failure is thus not a matter of degree but of which arm of immunity is engaged.

Ex vivo editing offers a third reference point that sharpens the comparison. In the CLIMB SCD-121 trial of ex vivo CRISPR-Cas9-edited CD34+ cells, 31 patients experienced no treatment-related serious adverse events and no anti-Cas9 antibodies were reported. The same source notes that AAV-delivered in vivo editing carries higher immunogenicity risk, with some patients requiring immunosuppression. Read together, these observations suggest that the relevant variable is not the nuclease but the context of its presentation: Cas9 delivered ex vivo, outside an immunologically active vector, elicits little detectable humoral response, whereas the same editing logic pursued in vivo via AAV does.

A comparative framework that quantifies these distinctions is available, though at coarse resolution. One cross-modality comparison characterizes four delivery routes along editing duration and safety: LNP editing is transient, rAAV is long-lasting, and electroporation is transient and highly invasive in vivo. On safety, both rAAV and LNP are judged potentially immunogenic, whereas electroporation is classified as nonimmunogenic but highly cytotoxic. This is a genuinely extractable contrast, and it is instructive precisely because it inverts the usual ranking: the modality with the least immunogenic liability carries the greatest mechanical cytotoxicity, so that "immunogenicity" and "safety" cannot be treated as a single axis.

The LNP case also reveals that innate and adaptive effects are not cleanly separable in outcome. In cynomolgus monkeys, PCSK9 editing efficiency reached only 25%, and anti-Cas9 antibodies were detected in blood, suggesting that T-cell-mediated killing of target hepatocytes limited efficacy. This is a notable complication for the "LNP avoids adaptive immunity" narrative: even where repeat dosing is tolerated, an adaptive response against the editing machinery can still erode durable editing. The failure mode here is coupled rather than singular—innate stimulation and anti-Cas9 humoral responses appear to interact

in constraining outcomes, which is precisely the kind of coupling that single-modality studies are poorly designed to detect.

RNP delivery, by contrast, is comparatively under-characterized in this evidence set. The ex vivo trial data provide indirect support for low immunogenicity of ribonucleoprotein-based editing, and the broader nuclease-immunogenicity literature treats transient nuclease exposure as favorable. However, no source in this collection reports a direct, quantified comparison of RNP against AAV or LNP in vivo. The absence is itself informative: the evidence base is dense for AAV, moderate for LNP, and thin for RNP, which means any claim that RNP is "least immunogenic" rests on mechanistic inference rather than measured contrast.

Taken together, the extractable evidence supports a two-axis characterization rather than a single immunogenicity ranking. Along the adaptive axis, AAV is clearly the most problematic, with pre-existing neutralizing antibodies and memory responses jointly blocking repeat dosing; along the innate axis, LNP carries the principal burden. Electroporation sits outside both axes, trading immunogenicity for cytotoxicity. This structure explains why cross-modality comparisons so often disagree: they are measuring different arms of immunity and reporting them under a single label.

The boundary conditions deserve emphasis. Most of the quantitative anchors—2–60% seroprevalence, 70% Serpinc1 knockdown, 25% PCSK9 editing—come from different species, tissues, and dose regimens, and none derives from a controlled head-to-head design. The apparent ordering of modalities is therefore an ordering of documented liabilities, not of measured immunogenicity under matched conditions. This is the same comparability gap identified at the level of seroprevalence measurement, now reappearing at the level of modality. What the literature supplies is a set of mechanistically coherent, partially quantified contrasts; what it does not supply is a common assay by which those contrasts could be placed on one scale.

6.3 Immunogenicity as a Multiplicative Attenuation Coefficient on Editing Persistence

Can immunogenicity be modelled as an attenuation factor on editing persistence, and can that factor be attached to the unified objective function established for the first research question? The evidence assembled here supports an affirmative but conditional answer: immune responses act multiplicatively on the durability of edited-cell populations rather than constituting a parallel risk axis, and the resulting coefficient can be given an operational form by borrowing the structure of existing composite efficacy metrics. The condition is that the coefficient must carry vector- and tissue-specific weights, because the same T-cell response that erases hepatic transgene expression leaves expression intact in immune-privileged compartments.

The case for a multiplicative rather than additive formulation rests on the mechanism by which adaptive immunity removes the therapeutic substrate. Capsid-specific CD8ˆ+ T cells recognize vector-derived epitopes presented through MHC class I cross-presentation on transduced hepatocytes and eliminate those cells directly, producing the characteristic elevation in serum transaminases accompanied by a decline in transgene expression [32]. This is not a competing hazard that runs alongside loss of editing; it is a process that destroys the edited-cell reservoir itself. Neutralizing antibodies act through an analogous logic on the extracellular route, binding vector particles and blocking the transduction events from which persistence would otherwise accrue [14]. Framing immunogenicity as a factor that scales down persistence therefore follows the biology more faithfully than treating it as an independent term summed into a composite risk score, since the two quantities are not exchangeable: a modest immune response against a large persistent reservoir may remove more edited cells than a strong response against a small one.

Clinical comparison across delivery modalities supplies the clearest quantitative hint that this factor is large enough to matter. In the lipid-nanoparticle trial reported by Rothgangl and colleagues, anti-Cas9 antibodies emerged yet editing efficiency reached only approximately 25%, whereas the immunologically quieter experiment of Musunuru and colleagues achieved more than 60% PCSK9 editing. The contrast is suggestive rather than decisive, since the two studies differ in dose, cargo and formulation as well as in immune readout, and no shared assay places the responses on one scale. Nevertheless the direction and magnitude are consistent with an attenuation coefficient well below unity in immunologically reactive settings, and they indicate that persistence estimates calibrated on immunologically silent experiments will systematically overstate durable editing in seropositive or previously exposed populations.

A usable prior for that coefficient is available, because pre-existing immunity is neither rare nor low-variance. Neutralizing antibodies against AAV2 are detectable in 40–74% of individuals and against AAV8 in roughly 40%, while anti-SpCas9 T-cell responses can be detected in 57–95% of healthy donors [14]. These seroprevalence figures matter for modelling because they define the population distribution from which a subject-specific attenuation term would be drawn, rather than a single scalar correction.

A unified objective that assumes an immunologically naive recipient is therefore not merely optimistic at the margin; it misrepresents the modal case. The same literature indicates that the magnitude of the attenuation depends on the pre-existing titre, which argues for a titre-conditional coefficient rather than a binary serostatus flag.

Dose dependence sharpens the functional form the coefficient should take. Host responses in the liver-directed setting have been observed principally at high intravenous doses, with doses above roughly $1.5 \times 10\hat{}14$ vg/kg associated with hepatotoxicity and renal injury [32]. This threshold behaviour implies that the attenuation term is not a constant but a function of dose, rising steeply once a tolerogenic ceiling is exceeded. A constant multiplicative penalty would therefore misprice low-dose regimens, which may retain most of their persistence, while under-penalizing high-dose regimens in which immune activation dominates the durability budget. The modelling implication is that the coefficient should be specified as a dose-dependent hazard-like term rather than a fixed discount.

The structural template for attaching such a term to the unified objective already exists in the clinical-translation literature. The Algorithm-to-Outcome Concordance metric combines a multiplicative efficacy term with an explicit uncertainty penalty, taking the form $(\mathrm{AUC}\times\mathrm{Corr})/(1+I\hat{}2/100)$. This decomposition is directly transferable: the efficacy product corresponds to the persistence estimate, and the uncertainty denominator corresponds to the confidence attached to that estimate, so an immune attenuation factor can enter as an additional multiplicative term without altering the architecture of the objective. The advantage of this placement is that attenuation propagates through every downstream utility calculation automatically, preserving the ordinal ranking of candidate designs under a single coherent scaling.

Converting the attenuated persistence into a clinical decision quantity requires a decision-theoretic interface, and the weighted Brier score supplies one. By assigning cost weights $w(c)$ to misclassification at threshold c, it decomposes discrimination from calibration and connects to net benefit, $\mathrm{TPR}(c) \cdot \pi - [c/(1-c)] \cdot \mathrm{FPR}(c) \cdot (1-\pi)$ [22]. An immune attenuation factor can thus be expressed as a shift in effective prevalence or as a cost weight on false reassurance, which is precisely what loss of persistence represents: a predicted durable response that fails in the clinic. This translation is what allows the coefficient to be optimized rather than merely reported, since it converts a biological phenomenon into a quantity with explicit clinical exchange rates.

The most important boundary condition is that the attenuation is not universal, and a coefficient fitted without regard to tissue will overcorrect. In muscle-directed, ocular and intrathecal delivery, T-cell responses have been documented without corresponding elimination of transgene expression [32]. The mechanistic reading is that immune privilege and local tolerogenic environments decouple detection of a response from functional clearance of edited cells, so the coefficient must be indexed by tissue and route. A single global attenuation constant would therefore penalize ocular and intrathecal strategies that in practice retain persistence, and would distort cross-modality comparison in exactly the direction the earlier analysis warned against.

Efforts to reduce the immunogenicity of the editing machinery itself offer a second lever, and one that acts on the coefficient rather than on the persistence estimate. Engineering of nucleases to remove immunodominant epitopes, and the use of base-editing architectures selected for minimized off-target activity and reduced immunogenic burden, both lower the probability that a response of consequence is mounted [39, 8]. In modelling terms these interventions shift the coefficient toward unity without changing the underlying target tissue, which makes them attractive because they improve durability without requiring dose escalation. The clinical-translation literature reaches a compatible conclusion, treating immunogenicity management as a design variable rather than a fixed property of the platform.

Taken together, the evidence supports modelling immunogenicity as a multiplicative, dose- and tissue-indexed attenuation coefficient on editing persistence, attachable to the unified objective through the multiplicative-efficacy and uncertainty-penalty structure already used in composite clinical metrics, with the weighted Brier score providing the route from attenuated persistence to net clinical benefit. The principal gap is not conceptual but metrological: the literature reports immune responses and editing outcomes in assays that cannot be placed on a common scale, so the coefficient can be specified structurally but not yet estimated with confidence across modalities. This unresolved calibration problem motivates the next stage of the argument, which turns from whether immunogenicity can be incorporated to what a reconstruction of the objective function would require in order to make such incorporation empirically accountable.

7 CHALLENGES IN INTEGRATING AND ENSURING COMPARABILITY OF MULTI-SOURCE OFF-TARGET BENCHMARKS

The accountability gap exposed by immunogenicity calibration further necessitates a reconstruction of the objective function itself, so as to explicitly specify the empirical constraints and structural conditions required when incorporating immunogenicity into optimization.

Having traced how neglected immunogenicity originates and how it might be reconstructed, the review now turns to the evidentiary substrate on which any safety claim must rest: multi-source off-target benchmarks. Their integration exposes a persistent tension—benchmarks proliferate across assays, cell types, and label thresholds, yet the topological structure of their relations remains star-shaped and radially divergent, with cross-benchmark comparison links too sparse to yield a genuine normalisation anchor. This chapter examines how such fragmentation propagates through dataset lineages and platform choices, why generic Accuracy, Recall, and Precision metrics fail successively at the levels of consequence, prior, and normative meaning, and whether a normalised risk scale can be defined at all—concluding that only protocol-bound, conditional scales are attainable, a limitation that motivates the quantitative translation framework developed next.

7.1 Topology of Benchmark–Metric Coupling and the Structural Gap in Cross-Benchmark Comparability

This section examines the distribution of associations between off-target detection benchmarks and evaluation metrics, and asks whether comparability links sufficient to support a unified risk scale already exist among different benchmarks. Synthesizing the available evidence, this association structure exhibits a pattern of "centralized radiation with sparse cross-benchmark bridging": a small number of benchmarks concentrate the vast majority of evaluation relationships, while the comparative relationships connecting different benchmarks, though identifiable, remain at the level of algorithmic juxtaposition and have not yet constituted a normalized scale skeleton.

The association distribution at the benchmark level is highly uneven. CHANGE-seq and GUIDE-seq are the only two off-target benchmarks that have attained the de facto status of "being jointly adopted by more than two independent studies," and around them converge more than a dozen evaluation relationships from a variety of methods, including DNABERT, XGBoost, One-Hot, K-mer, Bulges, NPM, and SHAP, forming the evaluation hub with the highest connection density in the entire repository. In contrast, the Yeast CAN1 assay is attached to only two evaluation relationships, the 13-technique compendium dataset to only one, and the Hanna dataset is likewise connected to only a single method[8]. This stark disparity in distribution is no accident: it reflects the path dependence of evaluation practice within the field—new methods tend to be validated on existing public benchmarks, whereas niche benchmarks, lacking reuse, find it difficult to accumulate comparable evidence.

However, hub status does not equal comparability. Two de facto standard benchmarks differ in scale by about 188-fold (CHANGE-seq covers 202,044 sites across 110 gRNAs, while GUIDE-seq covers approximately 1,072 sites across 58 gRNAs), yet AUPR, F1, Accuracy, and AUC are reported side by side in the same results table[41]. The scale difference alone is sufficient to alter the composition of positive and negative samples, and the practice of side-by-side reporting formally creates an appearance of comparability while substantively concealing distribution-level incommensurability. This suggests that the density of evaluation relationships and the transferability of evaluation conclusions are two mutually independent dimensions.

The more fundamental disagreement lies in label semantics. CHANGE-seq defines active sites by a machine threshold of cleavage count greater than 100 (67,476 active versus 2,806,152 inactive), whereas GUIDE-seq treats experimental detection as positive and uses Cas-OFFinder to generate negative samples (1,702 active versus 1,476,301 inactive)[41]. The two operational definitions of "what constitutes an active off-target" belong to different epistemological traditions: the former is a statistical cutoff on sequencing depth, while the latter is an existence judgment based on experimental capture. Any scheme that attempts to establish a unified risk scale between the two must first renegotiate label semantics rather than merely adjusting the numerical scale.

Differences in label semantics are further amplified by extreme class imbalance into a metric trap. The negative-sample proportion in both benchmarks is approximately 97%. Under this distribution, Accuracy can reach 0.990–0.992 and AUC can reach 0.978, whereas AUPR is only 0.345–0.612 and F1 is as low as 0.018–0.389[41]. On the same benchmark, the choice of metric alone can cause conclusions to differ by several times in magnitude. Some studies explicitly switch to AUPR on the grounds that AUROC incorporates true negatives on highly skewed data and thus creates an overly optimistic impression[41],

while other work reports an accuracy of 96.92% on an artificially balanced set. The heterogeneity of metric semantics causes cross-study numerical comparisons to lose focus from the very starting point.

Cross-benchmark comparison relations do indeed exist, but their nature warrants careful scrutiny. Several groups of benchmark- or method-level comparative associations can be identified across the literature, such as the contrasts between CCLMoff and CRISOT, and between DNABERT and CCLMoff [10]. The confidence of these associations is generally only 0.7, lower than the 0.95 for method-proposal relations, and their descriptions are mostly self-characterized as "implicit or empirical comparisons of alternative methods" rather than direct empirical contrasts or incremental inheritance. In other words, what they record is "the juxtaposition of different algorithms on the same task," not "the conversion between different benchmarks on the same scale."

The metric fragmentation of the same model across platforms provides a quantitative footnote to this. CCLMoff achieves AUROC 0.985 and AUPRC 0.720 on DIG-seq, and AUROC 0.944 and AUPRC 0.661 on DISCOVER-seq, whereas on GUIDE-seq the AUROC drops sharply to 0.279 while the AUPRC reaches 0.810[42]. This peculiar combination, in which AUROC and AUPRC point in opposite directions, indicates that the two metrics capture different ranking properties on this platform, and that a single metric cannot characterize the true risk profile of the model. This phenomenon is not an isolated case of model deficiency, but a direct projection of benchmark heterogeneity at the metric level.

Domain-level judgments corroborate the above observations. One review explicitly points out that existing off-target datasets cannot be meaningfully compared because they employ different detection methods, sensitivity thresholds, reference genomes, and variant-calling criteria, and that no deep learning model has yet demonstrated robust cross-platform generalization capability[33]. This assertion provides external support for the topological judgment of "connected but without a skeleton": the existence of connections does not imply the convertibility of scales, and the gap between the two can only be bridged through methodological reconstruction.

The endogenous bias of the benchmarks themselves further erodes the reliability of the hub benchmarks. Because of the limited delivery efficiency of dsODN, GUIDE-seq can detect only 30%–50% of all double-strand breaks, and its positive sample set is inherently incomplete, constituting a source of systematic false negatives[36]. At the same time, in vitro methods, being divorced from the intracellular chromatin environment, may overestimate off-target sites in real cells and introduce false positives[36, 43]. False negatives and false positives arise from two distinct pathways, capture efficiency and system simplification, respectively, yet in cross-benchmark comparisons they couple into a bidirectional scale shift, making the active sets of in vitro and intracellular benchmarks difficult to reconcile.

Coverage blind spots reveal another type of structural gap. For non-sequence-dependent off-target effects represented by base editing, their SNV off-target effects are considered to be independent of the target sequence and to arise from nonspecific random deamination; currently, they are covered only by a few systems such as Yeast CAN1, and the related evaluation relationships are extremely sparse[8]. This means that the existing evaluation scales, whose main axes are AUPR and Accuracy, cannot in principle be transferred to this modality. Although work in fields such as veterinary safety prediction and general medical imaging frameworks likewise faces the problem of incomparability among heterogeneous metrics, it has no direct connection to off-target benchmarks and can only serve as a meta-level analogical reference.

The relational structure examined in this section supports three judgments. First, evaluation relationships are highly concentrated on two de facto standard benchmarks, while the remaining benchmarks exhibit single-point dependence, resulting in a structurally skewed coverage. Second, although cross-benchmark comparative relationships are identifiable, they are predominantly implicit empirical contrasts lacking a unified measurement semantics and label definitions, and thus do not constitute normalized bridging. Third, the precondition for establishing a cross-benchmark risk scale lies in unifying the operational definition of active sites, replacing Accuracy and AUROC with imbalance-robust metrics, and incorporating non-sequence-dependent base editing modalities into the evaluation landscape. Until these conditions are met, any cross-benchmark numerical comparison rests on incommensurable labels and measures;

7.2 The Absence of Clinically Acceptable Semantics in General Classification Metrics

If cross-benchmark comparison already rests on incommensurable labels and metrics, a prior question must be asked: what exactly do Accuracy, Recall, and Precision fail to express when the target quantity is a clinically acceptable off-target rate? The evidence reviewed here indicates that the gap is not a matter of numerical precision but of semantics, and that it opens along three cumulative axes—consequence, prior, and normativity—each of which renders a different class of clinical information invisible to the metric.

The first gap is consequential: generic metrics weight every misclassification symmetrically, whereas

off-target safety assessment assigns radically asymmetric costs to the two error types. A false negative in this setting means an active off-target site is computationally cleared as safe, potentially converting a genotoxic event into an undetected one, while a false positive merely adds an experimental validation burden; the literature identifies the former as the most consequential error type in clinical safety assessment [33]. Accuracy and F1 are blind to this asymmetry by construction, since they aggregate errors without reference to downstream harm. The practical consequence is that a model can appear clinically reassuring while failing precisely where failure is least tolerable, and no reweighting of a single scalar can recover the lost distinction.

Empirical evidence bears out this masking effect. Under deliberately balanced conditions (472 active versus 472 inactive), tree-based classifiers report Accuracy of roughly 96.6–96.9% together with Precision near 96.8% and Recall near 96.3%, values that read as near-clinical-grade performance. Yet these figures are obtained on an artificially balanced sample and therefore say nothing about the operative difficulty—avoiding a handful of active sites among millions of inactive ones. Comparable evidence from a safety-evaluation framework outside genomics shows that a single evaluator can post high Accuracy while its false negative rate increases by at least 11.4% relative to a more robust protocol, confirming that high Accuracy and high miss rates can coexist because Accuracy is structurally insensitive to false negatives. This cross-domain analogy is methodological rather than direct, but it isolates the same mechanism.

The second gap is prior-dependent: extreme class imbalance decouples headline aggregate metrics from the discrimination that clinical use actually requires. CHANGE-seq comprises approximately 67,476 active against 2,806,152 inactive sites (about 97% negative), and GUIDE-seq comprises 1,702 active against 1,476,301 inactive [41]; integrated datasets span positive-to-negative ratios from 26:1 to 4189:1 [42]. Under such priors, Accuracy and AUROC are dominated by the majority class, and the divergence from precision-oriented measures becomes stark: on GUIDE-seq, Accuracy near 0.99 coexists with Precision as low as 0.009 for K-mer features and 0.066 for Bulges features [41]. A model described as 99% accurate may thus correctly rank fewer than one in ten of the positives that matter clinically.

The same dissociation appears in tool-level reporting. A language-model-based predictor reports Balanced Accuracy of 0.998 and AUROC of 0.985, yet F1 of only 0.409 and AUPRC of 0.524 on CIRCLE-seq [42]. The contrast between the two metric families is diagnostic rather than incidental: balanced accuracy and AUROC reward ranking across the full spectrum, whereas F1 and AUPRC expose behaviour in the rare-positive regime that corresponds to real genomic screening. Because the same Accuracy value carries entirely different meaning at a 26:1 and a 4189:1 prior, the metric is not merely optimistic but non-normalizable across benchmarks—a numerical identity that conceals a semantic difference.

A third, subtler mechanism compounds the first two: the highest-risk tail of the effect distribution is systematically compressed toward the mean. Error analysis of a feature-based predictor shows that for the worst decile of samples (true z-score below -4.0), the model consistently under-predicts effect magnitude and regresses toward the mean [41]. This is the region of greatest clinical concern, and it is precisely where aggregate metrics are least informative, since a handful of severely mispredicted tail cases barely moves Accuracy. The underestimation is therefore not random noise but a directional bias aligned against conservative safety estimation, and it operates independently of the imbalance problem described above.

The third gap is normative: clinical acceptability is defined relative to a specific indication, as whether the frequency, location, and type of unintended genomic events are acceptably low for that indication [33]. This is a multi-dimensional regulatory judgement, whereas Accuracy, Recall, and Precision are indication-blind descriptive statistics carrying no scale for consequence severity or acceptability. The mismatch is sharpened by the observation that large structural variants occur at low frequency yet carry genotoxic potential disproportionate to that frequency, so frequency-based metrics inherently undervalue low-frequency, high-consequence events; follow-up windows are in turn insufficient to exclude long-latency genotoxic events [33]. A metric that cannot represent latency or severity cannot represent acceptability.

A further dimension is missing altogether: editing efficiency and off-target risk are independent axes, and mitigation strategies frequently trade one against the other, as when nickase approaches afford lower editing efficiencies [43]. Clinical decisions require a joint acceptability scale over this efficiency–risk plane, yet the standard metric suite offers no mechanism for composing the two. Reporting a single off-target Precision or Recall therefore leaves the decision-relevant quantity—whether a given efficiency–risk pair is acceptable for an indication—entirely unrepresented, which is a gap of kind rather than of degree.

Taken together, these findings indicate a three-layer semantic deficit: a consequence gap (symmetric weighting versus asymmetric clinical cost), a prior gap (metric values drifting with imbalance and blocking cross-benchmark normalization), and a normative gap (absence of an indication-relative acceptability scale). The quantitative signature is consistent across sources—Accuracy near 0.99 alongside Precision of 0.009 [41], and Balanced Accuracy of 0.998 alongside F1 of 0.409 [42]. The convergence of independent

benchmarks on the same pattern strengthens the inference that the deficit is structural to the metric family rather than an artifact of any single study.

Two boundaries qualify this conclusion. First, the evidence establishes the mechanism but not its magnitude: no study in the reviewed literature quantifies the underestimation on a unified clinically acceptable off-target rate scale, and the insufficiency of follow-up to exclude low-frequency, long-latency events is disclosed qualitatively rather than numerically [33]. Second, the cross-domain evidence on Accuracy masking false negative rates originates in language-model safety evaluation and functions here as a methodological analogy, not direct proof. These limits do not weaken the diagnosis, but they do mean that closing the gap requires construction rather than correction.

The implication is that substituting one aggregate metric for another cannot bridge the divide. What is required is an asymmetric cost function that explicitly weights the clinical consequences of false negatives, imbalance-robust ranking measures such as AUPRC reported alongside their underlying priors, and an explicit, indication-relative acceptability threshold that maps an off-target rate onto clinical tolerability. Such a construction would supply the normalization scale that current benchmarks lack; how that scale might be operationalized across heterogeneous assays and label definitions is the subject to which the following section turns.

7.3 Protocol-Bound Normalization of Off-Target Risk

Whether a normalized off-target risk score, together with a clinically acceptable off-target threshold, can be defined so that results from different laboratories become comparable is the question this section addresses. The evidence assembled here supports a qualified answer: a normalization scale can indeed be constructed, but only as a protocol-bound relative scale, not as a universal absolute one. The obstacle lies less in model capability than in the measurement layer—how active off-target sites are defined, how negatives are generated, and how cleavage efficiencies are preprocessed. This section first examines why the prevailing metrics resist cross-laboratory conversion, then traces the irreducible trade-offs any scale must encode, and finally sets out the three-tier construction that the literature, taken together, makes plausible.

The dimensions the community actually measures are discriminative-performance indices rather than clinical risk quantities. Across the surveyed methods, Accuracy, Recall, Precision, and F1 are each independently reported by eight to ten approaches, whereas PR-AUC or AUPR appears in roughly four, and no dimension corresponds to a per-genome count of harmful events [10, 41]. This distribution is diagnostic: the field has converged on how well a classifier separates labeled sites, not on how much patient risk a score implies. Consequently, existing metrics can support normalization within a single study but cannot underwrite risk comparison across laboratories, because the quantity being normalized is discriminative capacity rather than hazard.

The most direct barrier is a divergence in positive-class definitions. CHANGE-seq labels sites with cleavage counts above 100 as active, yielding 67,476 active against 2,806,152 inactive sites, whereas GUIDE-seq treats all experimentally identified sites as active and generates negatives with Cas-OFFinder, producing 1,702 active against 1,476,301 inactive [41]. The prior positive rates therefore differ by roughly two orders of magnitude. Any uncalibrated probability output inherits this discrepancy, so that an identical score carries different empirical meaning under the two protocols; the incomparability originates in label semantics, not in the models trained upon them.

Feature representation introduces a second, subtler source of drift. The same XGBoost learner attains AUPR 0.661 and Pearson 0.756 on GUIDE-seq with One-Hot encoding, yet on CHANGE-seq its best configuration is Bulges encoding at AUPR 0.612 [41]. The optimal encoding thus shifts with the benchmark, which means a score axis calibrated under one representation does not transfer intact to another. DNABERT-Epi reinforces this point: it reaches PR-AUC 0.550 on GUIDE-seq, the strongest single model there, and pretraining adds 0.1653 PR-AUC over training from scratch ($p<0.001$), yet its authors state plainly that no single architecture is universally optimal across dataset conditions [10]. A universal optimum, if it existed, would supply a natural anchor for normalization; its absence leaves each benchmark to define its own reference point.

Preprocessing can silently rescale the target itself. One study reports Extra Trees accuracy of 0.9692, precision 0.9683, recall 0.9634, and F1 0.9668 on 472 targets and 16 sgRNAs, but only after applying linear regression to standardize cleavage efficiencies from two sources against the Tsai dataset as reference, the authors conceding that the original efficiencies are not comparable. The high scores are therefore relative to a constructed reference rather than to an absolute scale. Read alongside the encoding-dependent drift above, this indicates that cross-laboratory comparability fails first at the level of measurement definition and preprocessing, and only secondarily at the level of model capacity.

A further constraint is that normalization cannot be collapsed onto a single axis, because the underlying trade-offs are irreducible. Specificity and on-target efficacy move in opposite directions: high-fidelity variants and truncated sgRNAs raise specificity while depressing on-target activity [8, 33]. Recall and precision behave similarly under extreme imbalance; in a high-stakes safety classifier, SMOTE combined with ENN lifted minority-class recall from 0.64 to 0.86 at an explicit cost in precision [44]. Any scale that records only accuracy or F1 will therefore conceal the very risk it purports to summarize, since the CRISPR setting itself exhibits recall of 0.709–0.934 against precision of only 0.009–0.494 [41].

These trade-offs imply that a clinically acceptable threshold should be defined as a protocol-relative quantity rather than as a model score. The workable formulation is the upper bound on acceptable active off-target sites per genome at a given detection sensitivity, anchored to experimentally observable site counts [41]. Anchoring to counts rather than probabilities sidesteps the label-semantics problem described above, because a site count is defined by the assay protocol itself. The threshold then becomes comparable across laboratories precisely to the extent that the detection protocol is held fixed, which is a weaker but achievable form of comparability.

Calibration against a shared reference panel offers the mechanism for mapping arbitrary continuous scores onto such counts. Fitting an empirical positive-rate curve on a common sgRNA panel under a fixed detection protocol converts any score—whether a CRISOT-Score, a DNABERT probability, or an ensemble output—into the empirical frequency with which that score corresponds to an active off-target [42, 10]. A rudimentary anchor of this kind already exists: a CRISOT-Score above 0.8 almost always indicates an active off-target, and below 0.5 an inactive one [42]. That such thresholds are stated at all shows calibration is feasible; that they remain tied to a particular assay shows it is not yet portable.

Reporting conventions must likewise be two-axis, stating the on-target efficacy at which a given off-target reduction is achieved. Base-editing systems illustrate the pattern: AID-2S and AID-3S reduce R-loop off-target editing by approximately 4.5-fold and 13.7-fold relative to Target-AID while retaining appreciable on-target efficiency [8]. A single figure of fold reduction, reported without the accompanying on-target value, would be uninterpretable as a risk statement. The same logic applies to detection-side choices, where Digenome-seq resolves indels at frequencies of 0.1% or lower at high cost while CIRCLE-seq suffers a high false-positive rate [33, 43]; the measured risk shifts with the instrument, a dependency no single score resolves.

The reliability of any normalized score degrades precisely where clinical decisions matter most. Gradient-boosted regression systematically reverts toward the mean at extreme values, so that the worst decile of guides is underestimated [41]. A normalized score is thus inherently least trustworthy in the high-risk tail, which is the region a clinically acceptable threshold is meant to govern. This is not a defect of one implementation but a structural property of regression on rare high-effect events, and it argues for treating tail predictions as requiring orthogonal experimental confirmation rather than as calibrated risk estimates.

Two boundaries qualify the construction. First, no study in the assembled evidence has actually realized a cross-laboratory normalized scale; the works reviewed evaluate on multiple datasets but never construct a unified score axis [10, 41]. The claim that such a scale can be defined is therefore an inference from converging evidence, not a demonstrated result. Second, the recall–precision evidence borrowed from veterinary safety modeling operates on a different construct and cannot be transferred numerically; it supports the methodology of two-axis reporting, not any specific threshold.

Taken together, the evidence indicates that a normalized off-target risk scale is definable as a protocol-bound relative scale with three components: calibration of continuous scores against a shared reference panel, a clinically acceptable threshold expressed as an upper bound on observable active sites per genome, and mandatory two-axis reporting of off-target reduction against retained on-target efficacy [8, 42]. The two irreducible trade-offs—specificity against on-target efficacy, and recall against precision—dictate that no single-index scale can serve clinical decision-making. Standardization therefore lies in shared reference panels, fixed negative-generation rules, and count-anchored thresholds, not in more elaborate models; how such protocol-bound scales might be reconciled when assays themselves disagree remains the open problem.

8 A QUANTITATIVE TRANSLATION FRAMEWORK FOR QUALITATIVE LIMITATIONS AND ASSUMPTIONS

After the shared reference panel, the fixed negative-generation rule, and the count-anchored threshold jointly define the standardization boundary, how to reconcile these protocol-constrained scales when divergence arises among assays requires shifting from the setting of standardization conditions to the systematic

evaluation of cross-assay consistency.

Having established why multi-source off-target benchmarks resist direct comparison, the review now turns from diagnosis to formalization: the qualitative limitations and assumptions documented across the evidence base are not residual caveats but structured, quantifiable constraints that can be carried into a risk model. The central tension is that these constraints concentrate along a small number of lifecycle and evidence dimensions yet remain disconnected from the probabilistic machinery that would make them actionable, so their translation is partial rather than complete. Accordingly, the chapter first maps the dominant axes of limitation and assumption, then identifies which assumptions should enter as conditioning variables and why current linkages leave them inert, and finally assesses how far existing clinical-translation roadmaps can be structured into risk-matrix and Bayesian-network form. This trajectory sets up the transfer of uncertainty-modeling practice from LLM safety to CRISPR safety, where the same conditional structure must ultimately be made explicit.

8.1 Dimension Distribution of Qualitative Limitations and Their Transformation into Risk-Model Inputs

A central question for any quantitative translation framework is where qualitative limitations actually concentrate, and which of them can be converted into variables that a risk model can consume. The structured evidence assembled here indicates that limitations are not scattered uniformly across the research space but cluster around validation and deployment, and that this clustering determines which transformation strategies are feasible.

The distribution of limitation statements is markedly skewed toward the validation and deployment stage, and within that stage toward conditions of uncertain evidential value. Typical entries in this region concern training-data limitations, cross-platform generalisability, the absence of external validation in primary studies, limited prospective outcome evidence, and heterogeneity across study populations [33]. This is a consequential pattern: the limitations that authors most often record are precisely those that bear on whether a claim will hold outside its original setting, rather than on whether the underlying mechanism is plausible. The concentration suggests that the dominant epistemic weakness in this literature is not explanatory but translational.

Two broad transformation routes can be distinguished, and they differ in what they require of the source statement. The first treats a limitation as a distribution over an outcome and propagates it through simulation; the second treats it as a position on a structured ordinal scale and reads off a discrete level. The first route is exemplified by work that assigns each qualitative risk a probability density function—deterministic risks as Bernoulli, random risks as triangular or normal, epistemic uncertainty as uniform—and then ranks absolute impact via Monte Carlo simulation and P95 percentiles. This approach yields a computable ordering from purely qualitative input, but its own stated dependencies are instructive: results hinge on expert elicitation of the distributions, and the model assumes risks are mutually independent. That independence assumption is exactly what a richer dependency structure would have to replace, which marks a clear boundary on direct reuse.

The second route is visible in engineering safety, where an intermediate variable set is already standardised. Risk-graph methods map a hazard onto severity, frequency of exposure, and possibility of avoidance, and combine these into a required performance level; product-side practice instead multiplies severity, occurrence, and detectability into a risk priority number. The significance of these triplets for present purposes is that they accept ordinal judgements as input and never require a direct probability estimate, which makes them well suited to safety-type limitations whose source literature reports categories rather than rates. Compared with the simulation route, this route is more robust to sparse evidence but coarser in resolution.

A more demanding variant constructs an explicit chain from hazard analysis through fault trees to basic-event probabilities, and proposes estimating the misclassification probability of a learned component as a component-level reliability claim, thereby decomposing a system-level safety target into computable requirements on individual parts. The contrast with the risk-graph approach is one of granularity versus tractability: the chain supports quantitative decomposition but presupposes that component failure statistics exist, whereas the ordinal triplet tolerates their absence at the cost of resolution.

Not all limitations are equally tractable, and immunogenicity provides the clearest counter-case. Anti-drug-antibody measurement is not quantitative because no standardised human polyclonal reference reagent exists, and observed rates shift with assay sensitivity, sample handling, and sampling time, to the point that cross-study comparison is described as not yet possible. Immunological monitoring for viral vectors faces the same deficit, compounded by the infeasibility of collecting relevant samples [32]. These limitations are protocol-absence limitations rather than knowledge-absence limitations: no amount

of modelling resolves them until the measurement procedure is fixed, so they must be excluded from the input set or carried as categorical flags.

Where a measurement protocol does exist, the same class of limitation becomes quantitatively expressible. Delivery-associated off-target risk can be calibrated by carrier type: a synthesis of sixty-seven studies reports typical off-target rates below one percent for ribonucleoprotein delivery, five to fifteen percent for plasmid, and roughly a fivefold reduction for mRNA relative to plasmid [31]. This establishes a quantifiable mapping from a categorical design choice to a risk mean, which is directly usable as a model input. The mechanism is instructive—the carrier determines the duration and level of nuclease exposure, so the limitation is mediated by an identifiable physical quantity rather than by study artefacts.

Immunological cross-reactivity admits a comparable treatment through multi-stage scoring. A prediction platform filters candidate peptides by processing likelihood, binding affinity below a threshold, and presentation probability, then ranks them by T-cell receptor-facing residue similarity, single-nucleotide variation, and expression level, converting a discrete catastrophic outcome into an ordinal safety score. The contrast with the assay-standardisation case is sharp: here the obstacle is not missing measurement but the combinatorial space of possible cross-reactions, and staged computational filtering substitutes for exhaustive experimental screening.

The clinical risk-prediction literature supplies the most direct evidence on what happens downstream once limitations are converted. Traditional models for spinal surgery achieve areas under the curve of roughly 0.64 to 0.78, machine-learning models improve discrimination by only about 0.05 to 0.10, external validation commonly reveals calibration drift, and only four of twenty-three primary studies were at low risk of bias across all four assessed domains. The failure mode is therefore not the absence of a quantitative model but the non-transferability of its inputs: variables that cannot be calibrated or externally validated propagate their defects into every downstream prediction. This is the same weakness that the validation-stage clustering of limitation statements had already anticipated.

Taken together, the evidence supports a three-tier classification of limitation statements by transformation readiness. Limitations with an established measurement protocol and a known mediating quantity—carrier-dependent off-target rates, staged immunological scoring—can be assigned numeric distributions directly. Limitations with ordinal but reliable expert judgement—safety hazards—can enter through severity, exposure, and avoidability triplets. Limitations whose measurement protocol is itself contested cannot be quantified and should be retained as categorical constraints. The distributional evidence indicates that the first tier is a minority and the third is concentrated in the biological and clinical domains, which bounds how much of the qualitative record a purely probabilistic model can absorb. The following section therefore turns from classification to the dependency structure that any such model must specify.

8.2 Conditional Variables for Risk Models: Which Assumptions Qualify and Why They Remain Unlinked

Which assumptions in the qualitative record genuinely qualify as conditional variables of a computable risk model, and why does the current evidence base fail to connect them, is the question this section addresses. The preceding classification established that only a minority of recorded assumptions carry the structural features a probabilistic model can absorb; the task now is to specify the dependency structure such a model would require and to diagnose the gap between that requirement and what the literature actually supplies. Three candidates named in the corpus—chromatin accessibility, context-dependent delivery, and the claim that cargo choice determines the delivery window—serve as the analytical thread, because each is simultaneously a well-attested biological modulator and a structurally isolated statement in the assembled record.

The formal definition of a Bayesian network settles the membership question before any empirical argument is needed. A Bayesian network encodes the conditional dependence structure among random variables through a directed acyclic graph, with conditional probability tables typically elicited from expert knowledge. Any quantity that modulates the probability of a downstream event and can itself take different values therefore belongs in the graph as a parent node, irrespective of whether a given study has named it as such. Chromatin accessibility satisfies this criterion exactly: heterochromatic compaction restricts Cas9 binding and reshapes the off-target profile, so the variable changes the likelihood of the adverse event while admitting distinct states across cell types and loci. The same reasoning applies to cargo choice, which shifts off-target indel frequencies by more than an order of magnitude and thus cannot be treated as a fixed modelling constant.

Empirical evidence converts this definitional argument into a quantitative one. GUIDE-seq off-target sites are significantly enriched in open chromatin as measured by ATAC-seq, and SHAP analysis of DNABERT-Epi identifies H3K27ac, an active-enhancer mark, as the single strongest feature for predict-

ing intracellular off-target activity [10]. Independent review evidence reaches the same conclusion from the opposite direction, noting that chromatin state is neglected in Digenome-seq assessment and must be compensated by DIG-seq, which directly confirms that chromatin status conditions off-target activity [36]. These two lines of evidence differ in method—one predictive and feature-attributional, the other assay-comparative—yet converge on the same causal direction, which strengthens the case that chromatin accessibility is a genuine parent variable rather than a correlated covariate.

Cargo choice exhibits an even sharper quantitative signature. Lipid- and polymer-based delivery of Cas9 ribonucleoproteins typically yields off-target indel frequencies below one percent, whereas plasmid delivery produces five to fifteen percent; the source states explicitly that cargo choice directly determines the acceptable delivery window, since sustained plasmid expression raises off-target risk while transient RNP delivery minimizes it [31]. Context-dependent delivery is formalized in the same review as a three-part decision matrix over target-organ accessibility, cargo size and desired expression duration, and acceptable safety risk [31]. This is a notable contrast with chromatin accessibility: the delivery variable has already been written as an explicit decision parameter in the primary literature, whereas chromatin state remains a predictive feature that no therapeutic decision framework has yet promoted to a formal modelling variable.

A workable quantitative container for such variables already exists in the process-safety literature. Layer-of-Protection Analysis computes consequence frequency as P consequence $=$ IEF $\prod$ PFD_i, and the same framework extends to credal networks that express data uncertainty as lower and upper probability bounds rather than point values, explicitly substituting a range for a fixed number [45]. This interval representation suits assumptions such as chromatin accessibility, which have a well-defined direction of effect but no reliable point estimate. The safety-integrity framework for automated driving supplies the complementary structural rule: additional nodes should be introduced to represent common-cause and coupling factors, and every edge should run from cause to effect. Chromatin accessibility and cargo choice are precisely such coupling factors, shared across targets and delivery modalities, and by this rule they belong at the top of the graph rather than as modifiers buried in a conditional table.

Engineering safety demonstrates that parameterizing assumptions is not merely feasible but routine. The ISO 13849 risk graph derives a required performance level from three parameters—severity, frequency of exposure, and possibility of avoidance—so that each qualitative judgement becomes an explicit input to a quantitative safety calculation. The contrast with gene therapy is instructive: in machine safety the parameterized assumption is the standard modelling primitive, whereas in the CRISPR literature the equivalent assumptions remain prose statements in review articles. This asymmetry is not a difference in scientific maturity but a difference in schema, and it explains why the same conceptual object appears in two disconnected forms across the corpus.

That disconnection is visible as a duplication of concepts. The problem statement ”chromatin accessibility affecting off-target profiles” is linked to a dedicated review through an explicit addresses relation with confidence 0.85 [33], while its assumption-side twin carries no connections at all. The same pattern recurs for delivery: the problem of lacking an efficient and safe delivery system is connected to a polymer-based non-viral delivery study at confidence 0.95 [5], yet the corresponding context-dependent delivery assumption is isolated. Two copies of one concept thus coexist, one richly attested and one orphaned, with no bridging relation asserting that the assumption instantiates or conditions the problem. The isolation is therefore an artefact of how statements were recorded, not evidence that the underlying knowledge is absent.

A third structural cause is the absence of a cross-domain schema. Off-target prediction tools such as DNABERT-Epi [10] and the broader off-target literature [36] inhabit a computational-biology subgraph, while Bayesian-network and LOPA methodologies [45] and safety-integrity frameworks [17] inhabit a process-safety subgraph. No method in the record simultaneously targets an off-target problem and requires a delivery or chromatin assumption, so there is no natural attachment point for the conditional variable. The gap is one of connective vocabulary rather than of missing evidence, and it would persist even if every relevant primary study were added to the corpus.

The methodological tensions in the available machinery must be acknowledged rather than smoothed over. Probability–impact matrices are known to distort quantification, sometimes assigning similar ratings to values that differ by orders of magnitude, and Monte Carlo simulation as commonly applied assumes independent risks [21]—an assumption directly at odds with the shared causal dependence among chromatin state, cargo choice, and delivery route. Even where a network is constructed, its probabilities may still derive from human judgement, and imprecise intervals may support only comparative rather than absolute conclusions [45]. Connecting the variables therefore removes a structural defect without resolving the deeper problem of parameter provenance.

The strength of the claim that these assumptions ought to be conditional variables should be calibrated

honestly. No study in the corpus explicitly names chromatin accessibility as a node in a Bayesian network; the safety-integrity framework supplies only the general principle that coupling factors become nodes, without applying it to genome editing [17]. The conclusion therefore rests on structural inference from the definition of conditional dependence [45] combined with convergent empirical evidence on effect direction [10], and should be read as a well-supported inference rather than a directly documented finding. This distinction matters for how the resulting model should be validated.

Related work outside the CRISPR domain reinforces both the feasibility and the caution. Risk prediction models in spine surgery illustrate the difficulty of translating a statistically adequate model into clinical use, where calibration and prospective validation rather than discrimination alone determine adoption. Safety arguments for machine learning similarly treat reliability assessment as an evidential activity that must accompany, not replace, the quantitative model [19]. Immunogenicity of AAV vectors and design strategies to reduce it supply a further example of a context-dependent hazard whose probability varies with vector, dose, and host state [32]„ suggesting that the conditional-variable logic generalizes beyond the three candidates examined here.

Taken together, the assumptions that qualify as conditional variables are those that modulate downstream probability, admit distinguishable values, and stand in a clear causal relation to the adverse event: chromatin accessibility as a parent of off-target activity, cargo choice as the determinant of the delivery window, delivery context as a tissue- and cargo-conditioned factor, and vector capacity and expression duration as threshold and exposure variables respectively. Their present isolation reflects duplicated concept recording, missing bridging relations, and the absence of a shared schema between the computational and safety-modelling literatures, compounded by genuine uncertainty about parameter values. The next section examines how such variables, once admitted, can be assigned probabilities that remain defensible from preclinical evidence to clinical decision.

8.3 Structuring Roadmap Risk Elements into Matrix and Bayesian Network Representations

Can the risk elements embedded in existing clinical translation roadmaps be structured into the nodes and conditional probabilities of a risk matrix or a Bayesian network? The preceding analysis established that the principal obstacles to quantitative translation are conceptual recording gaps, missing bridging relations, and the absence of a shared schema; the present section asks whether the roadmap literature nonetheless supplies risk elements concrete enough to populate such a formalism, and where that population remains underdetermined. The evidence suggests a qualified yes at the level of variable specification, coupled with a clear mechanism gap at the level of probability assignment.

The most direct contribution of the clinical roadmap literature is a compact, three-part decision matrix. A reanalysis of sixty-seven studies proposes that therapeutic CRISPR-Cas9 delivery choices be organised along three axes—target-organ accessibility, cargo size and expression duration, and acceptable safety risk—which together constitute a decision structure that maps almost one-to-one onto the cells of a conventional risk matrix [31]. This is significant because the axes are not merely categorical: each admits ordinal or continuous grading, so the matrix is populated rather than merely partitioned. The same synthesis supplies explicit factor values, reporting ex vivo editing efficiencies frequently above 80% while in vivo efficiencies in non-hepatic tissues remain below 30% for most non-viral systems, and contrasting RNP off-target rates below 1% with plasmid-associated rates of 5–15% [31]. These figures are precisely the kind of influence-factor variables that a Bayesian network requires at its input nodes, and their availability in a single comparative source distinguishes the delivery roadmap from more discursive guidance documents.

Off-target risk, however, resists reduction to a single axis and must be decomposed before it can serve as a severity dimension. A structured treatment of off-target effects separates seed-region mismatch tolerance, chromatin accessibility, and the sensitivity of structural-variant detection into distinct dimensions, each with its own measurement logic [33]. This decomposition is analytically useful for a risk matrix, since severity can then be conditioned on the detection modality rather than treated as an intrinsic property of the edit. Yet the same source is explicit that detection workflows lack standardisation, so that reported rates are not comparable across studies [33]. The consequence for formalisation is direct: a node labelled "off-target severity" can be defined, but its conditional probabilities cannot be pooled from the literature without an explicit correction for assay sensitivity. This is a boundary condition on the mapping, not a refutation of it.

A working precedent for automating exactly this kind of mapping comes from outside medicine. The SQUADfps framework converts qualitative hazard descriptions into structured scores by combining an ISO 13849 risk graph, in which severity, frequency, and probability of avoidance jointly determine the required performance level, with a process-FMEA in which the risk priority number is the product of severity,

occurrence, and detection. The relevance to clinical roadmaps is that both instruments accept the same qualitative inputs that roadmap authors already provide, and both output a scalar that can be ranked and thresholded. In contrast to the medical literature, where risk grading is typically narrative, this framework demonstrates that the translation from prose hazard to computable score can itself be formalised and audited.

The mechanism gap between a populated matrix and a probabilistic model is filled by work in process-safety quantitative risk analysis. There, a Bayesian network is defined as a directed acyclic graph grounded in Bayes' theorem, with conditional probability tables typically elicited from expert knowledge, and consequence frequency is quantified through a layer-of-protection formula in which the initiating event frequency is multiplied by the product of the probabilities of failure on demand of each independent protection layer [45]. The same treatment introduces credal networks, which represent imprecise probabilities as intervals rather than point values [45]. This is the component most directly transferable to clinical translation, because the uncertainty surrounding, for example, long-term off-target consequences is better characterised by an interval than by a single elicited number.

That roadmap nodes can indeed be assigned conditional probabilities and organised as a graph is demonstrated concretely in the automated-driving safety literature. A safety integrity framework there adopts a standard skeleton in which a scenario leads to hazardous behaviour, which leads to a collision, which leads to injury, and quantifies risk as the product of scenario frequency, the probability of hazardous behaviour, the probability of collision given that behaviour, and the probability of injury given collision [17]. The structural correspondence with the clinical case is close: target-organ accessibility plays the role of scenario exposure, editing efficiency and delivery fidelity the role of hazardous behaviour, off-target modification the role of collision, and clinical consequence the role of injury. What the clinical roadmaps supply, and what this framework assumes, is the numerical content of the intermediate terms.

A second, independent instantiation reinforces the feasibility of the approach while exposing a difference in maturity. A quantitative safety-risk assessment method for vehicular networks likewise decomposes system risk into conditional stages and assigns probabilities at each, yielding a composite measure suitable for comparison across design alternatives. Read alongside the automated-driving framework, the two establish that staged conditional decomposition is a stable design pattern rather than a one-off construction. The contrast with the clinical literature is one of evidentiary status: in the engineering cases the conditional probabilities are supported by field or simulation data, whereas in the clinical case they would rest on heterogeneous preclinical studies whose comparability is itself contested [33].

The matrix formalism itself, however, carries known limitations that any clinical adaptation must confront. A sustained critique of probability-impact matrices identifies two structural defects: they cannot represent dependencies among risks, and their probability and impact estimates are imprecise [21]. The proposed remedy is to replace the matrix with Monte Carlo simulation of each risk's absolute effect on schedule and cost, producing distributions rather than point rankings [21]. Transposed to clinical translation, this argues that roadmap limitations such as immunogenicity or delivery inefficiency should be modelled as distributions over outcome magnitudes, not as single cells in a grid—a point that bears directly on how the conditional probability tables discussed above ought to be elicited.

The dependency problem is not hypothetical in the gene-editing context. Immunogenicity of recombinant AAV vectors is modulated by capsid design, dose, and pre-existing serology, and design strategies intended to reduce immunogenicity can alter transduction efficiency in ways that feed back into the efficacy term [32]. Parallel work on immunogenicity-by-design makes the same coupling explicit, treating immunogenic potential as a design variable rather than an independent failure mode. A risk matrix that assigns immunogenicity and efficacy to separate, non-interacting cells therefore misrepresents the underlying structure; a directed graph in which immunogenicity is a parent of effective dose is the more faithful representation. This is the clearest instance in which the network formalism outperforms the matrix, and it follows from the biology rather than from any preference for one formalism over another.

Detection sensitivity introduces a further layer that the engineering precedents handle explicitly but the clinical roadmaps do not. In the process-safety treatment, protection layers are modelled as independent barriers with their own probabilities of failure on demand, so that the reliability of detection is itself a modelled quantity [45]. The clinical analogue—the sensitivity of the assay used to declare an edit off-target-free—is acknowledged as a source of incomparability [33] but is not represented as a node in any roadmap reviewed here. The gap is therefore specific and remediable: it concerns the absence of a detection-reliability term, not the absence of a formalism capable of accommodating one.

Structured risk taxonomies from the safety-evaluation literature offer a template for closing part of this gap. SafeBench organises safety risk into eight major categories and twenty-three subcategories and evaluates systems with two complementary quantitative indicators, a safety risk index and an attack

success rate. The dual-indicator design is instructive because it separates the magnitude of realised harm from the frequency with which protective behaviour fails, mirroring the severity and occurrence axes of a conventional matrix while keeping them numerically distinct. A clinical roadmap could adopt an analogous pairing, reporting consequence severity and the rate at which preclinical assays fail to detect a given hazard as separate quantities.

Prediction tools developed for editing specificity illustrate both the promise and the fragility of populating the probability terms. Sequence-based off-target prediction has advanced through language-model representations that improve discrimination of true editing sites from background [10], and dedicated platforms now predict the potential off-target binding of T-cell receptors [46]. These tools supply the kind of calibrated likelihood that a conditional probability table would require. However, reviews of off-target effects in CRISPR-Cas9 editing continue to document substantial variation in reported rates arising from assay and analysis choices [36], and base-editing systems with minimised off-target activity show that the underlying rates are themselves engineering-dependent [8]. The probability assigned to an off-target node is therefore conditional on the editor, the assay, and the analysis pipeline, and any network must carry these as explicit conditioning variables.

Non-viral delivery provides a useful test case for whether the roadmap's factor values survive contact with a network formulation. Polymer-based delivery of Cas9 complexes achieves targeted editing while avoiding the immunogenicity concerns associated with viral vectors, but its efficiency in non-hepatic tissues remains below the thresholds that the delivery roadmap identifies for therapeutic viability [5]. In network terms, this is a case where the efficacy node and the immunogenicity node move in opposite directions as a function of the same design choice, producing a trade-off surface rather than a single operating point. The roadmap reports the endpoints of this trade-off; the network would represent its shape.

The translation of these formalisms into clinical decision-making is itself an object of study, and the evidence there counsels restraint. A scoping review of risk prediction in spine surgery finds that traditional models and artificial-intelligence approaches alike face persistent difficulties in clinical translation, with calibration and generalisability proving more consequential than discrimination. This finding bears directly on the present question: a Bayesian network whose conditional probabilities are well specified but poorly calibrated will not support clinical decisions, and the clinical-prediction literature suggests that calibration is where such models most often fail. Reliability assessment and safety-argument work for machine-learning components makes a compatible point, namely that the assurance case must address the model's behaviour under distribution shift rather than only its average performance [19].

Taken together, the evidence supports a differentiated answer. The roadmap literature supplies a three-axis decision structure and a set of factor values—editing efficiencies, off-target rates by cargo format, and organ-specific delivery ceilings—that are directly usable as network nodes [31], while the off-target literature supplies the dimensional decomposition needed for severity grading [33]. The engineering literature supplies the missing machinery: staged conditional decomposition, protection-layer quantification, and interval-valued probabilities for imprecise parameters [17, 45]. What remains unresolved is the calibration and comparability of the probability terms themselves, a limitation that the matrix critique anticipates and that the clinical-prediction literature confirms as the principal barrier to translation [21]. The mapping is therefore feasible in structure and partial in content, and the residual indeterminacy is concentrated precisely where preclinical evidence is least commensurable.

9 TRANSFERRING LLM UNCERTAINTY MODELING TO CRISPR SAFETY

Now that structural feasibility has been established, and content-level incompleteness is concentrated precisely where preclinical evidence is least commensurable, the task that follows is no longer to argue whether the mapping can hold, but to examine how this partial mapping organizes comparable levels of evidence under specific themes.

Building on the preceding translation framework, this chapter examines whether the uncertainty-modeling apparatus developed for large language models can be carried into CRISPR safety assessment. The central tension is that the transferable elements are mechanistic and methodological rather than procedural: attack success rate, aleatoric and epistemic uncertainty, bootstrap and Δ-method propagation, and the recognition that aggregate benchmark scores are comparative rather than absolute each offer a disciplined vocabulary, yet no integrated pipeline has been validated in the CRISPR setting, and the evidence base contains no genuine cross-cluster precedent for such transfer. The chapter therefore proceeds from what can be directly transferred, through the separation and propagation of aleatoric from epistemic uncertainty, to the threshold-calibration schemes that would be required to support regulatory communication.

9.1 Mapping Established LLM Safety Metrics onto CRISPR Off-Target Assessment

Which of the safety metrics and conceptual distinctions matured in the LLM literature—attack success rate, the aleatoric/epistemic uncertainty dichotomy, bootstrap and Δ-method uncertainty analysis, and the principle that aggregate benchmark scores are comparative rather than universal safety thresholds—admit a faithful counterpart in CRISPR safety evaluation? The mapping is not merely terminological: each construct carries an implicit theory of what can go wrong and how failure should be counted, and the value of the transfer lies in whether that theory survives the change of domain. We examine the four constructs in turn, identifying where the correspondence is exact, where it requires inversion, and where the CRISPR evidence base simply lacks the measurements needed to instantiate the concept.

The aleatoric/epistemic dichotomy transfers with unusual precision, because CRISPR off-target assessment already separates two qualitatively different sources of uncertainty. Aleatoric uncertainty corresponds to the irreducible stochasticity of off-target cleavage itself—seed-region thermodynamics and chromatin accessibility jointly modulate whether a near-cognate site is cut, and this variability persists no matter how much data are collected [33]. Epistemic uncertainty, by contrast, corresponds to ignorance of undetected off-target sites and structural variants, which manifests empirically as failure to generalise across platforms. This framing mirrors the safety-integrity decomposition developed for automated driving, where intrinsic sensor noise is separated from specification incompleteness [17], and the operational-range bounding used in spectroscopy, where a model's validated envelope is distinguished from the region beyond it [47]. The correspondence is therefore structural rather than analogical: both domains require that reducible ignorance be reduced by evidence collection while irreducible noise be bounded by design margins.

Bootstrap and Δ-method estimators, by contrast, are transferable without loss, since they operate on the statistical properties of an estimate rather than on domain semantics. A worked protocol for trial-level resampling—10,000 bootstrap replicates with 2.5th–97.5th percentile confidence intervals, combined with a sensitivity analysis sweeping σ Corr from 0.05 to 0.20 within an $AOC = AUC \times Corr \times (1 - I\hat{}2)$ composite—provides a directly reusable template for propagating uncertainty through a composite safety metric [48]. An equivalent construction based on kernel-density bootstrap for estimating the probability of missed injury demonstrates that the approach extends to tail-oriented quantities of the kind CRISPR safety actually cares about [19]. Because neither estimator presupposes a particular data-generating process, they can be applied to off-target rate estimates without modification; what must be re-derived is only the quantity being resampled.

Attack success rate, however, does not transfer directly and must be inverted. In its native form ASR counts the percentage of adversarial inputs that cross a fixed ordinal threshold, an operationalisation used to score jailbreaking and privacy leakage in medical AI and agentic safety risk in deployment settings [49]. The CRISPR analogue is a thresholded off-target editing rate, but the direction of concern is reversed: the danger lies not in attacks that succeed but in edits that escape detection. This inversion is forced by quantitative evidence that thresholded summaries are insensitive to false negatives—a model reporting AUROC of 0.9683 while achieving AUPRC of only 0.51 under class imbalance shows that high aggregate discrimination can coexist with poor recovery of the rare positive class. Consequently, a migrated metric should be paired with precision-recall style measures rather than reported as a single success proportion, and the threshold itself should be treated as a design parameter subject to sensitivity analysis [10].

The principle that aggregate scores are comparative rather than universal safety thresholds transfers cleanly and, in doing so, exposes the central blind spot of CRISPR evaluation. A benchmark score of 85.27 on a clinical-safety-oriented evaluation establishes relative standing among systems, not an absolute guarantee of safety, a point reinforced by methodological work arguing that safety benchmarks measure what they operationalise rather than safety itself [50]. The CRISPR literature reaches the same conclusion from the opposite direction: there is currently no unified safe threshold for off-target rates, and the characterisation framework applied to a approved therapeutic functions as a case-specific reference standard rather than a generalisable cutoff [33]. The convergence is instructive—both fields possess ranking instruments while lacking the calibration evidence that would license a pass/fail interpretation.

Cross-platform generalisation failure is the CRISPR expression of benchmark incomparability, and it is more severe than in the LLM setting. No current deep learning model has demonstrated robust cross-platform generalisation, and comparative experiments across datasets show that model rankings can invert when the evaluation dataset changes [51, 10]. Reported performance is further conditioned on narrow experimental scope: one genome-wide prediction system validates only under NGG PAM and bulge-free conditions, so its headline figures describe a restricted operating regime rather than general editing behaviour [9]. This is the CRISPR counterpart of the K-Bench problem—a score that is meaningful only relative to the distribution on which it was computed—and it implies that any transferred evaluation

protocol must report the platform, PAM constraint, and cell context alongside the metric.

A further gap concerns stratification. No reviewed CRISPR study reports off-target rates disaggregated by cell type, tissue, or patient subgroup, which means the field has no equivalent of the subgroup reporting that clinical risk prediction has only recently begun to demand; a scoping review of surgical risk models found that no published studies reported performance across sociodemographic categories, which its authors identify as a critical equity gap. The same absence in genome editing is documented in the off-target literature [33]. Transferring LLM safety metrics therefore cannot by itself close this gap: fairness-oriented evaluation constructs have no CRISPR measurement to attach to, and the limitation is one of data collection rather than of metric design.

Experimental evidence illustrates how entangled the two uncertainty types become at the single-cell level, and why the dichotomy should be treated as an analytical convenience rather than an ontological partition. In edited lines, fewer than five predicted off-target sites carried mutations while five to seventeen non-predicted sites did, with no overlap with cancer genes; yet one line exhibited trisomy of chromosome 7, attributed to culture-induced aneuploidy rather than to editing. The trisomy is aleatoric with respect to the editing process but would be epistemic with respect to a model trained only on predicted sites, since the model never represented it. This suggests that assigning an observed anomaly to one category depends on the scope of the predictor, and that safety arguments must therefore document both the noise model and the validated envelope.

Taken together, the four constructs map onto CRISPR safety assessment with differing fidelity: the uncertainty dichotomy and resampling estimators transfer essentially intact, attack success rate transfers only under inversion and with precision-recall companions, and the comparative-not-universal principle transfers as a caution that the field already exemplifies. What does not transfer is the measurement infrastructure—stratified reporting and cross-platform calibration—without which the migrated metrics remain unanchored. These findings establish the conceptual vocabulary for the transfer; the following section turns to the mechanisms by which uncertainty estimated in silico can be propagated into actionable safety margins for editing protocols.

9.2 Cross-Domain Precedent for Sequence-to-Risk Prediction: Evidence, Architecture, and the Missing Uncertainty Layer

Can a language model, given sequence, vector, and dose inputs, predict off-target and immune risk, and does the existing literature already contain any cross-domain precedent linking LLM safety evaluation to CRISPR risk prediction? The evidence assembled here answers the first half affirmatively for off-target risk and only partially for immune risk, while the second half reveals a structural absence: no documented study connects the two research clusters through an explicit extension, comparison, or support relation. What exists instead is a set of adjacent precedents whose architecture is transferable even though their evaluation apparatus is not.

Sequence-based off-target prediction is, first of all, an established supervised learning problem rather than an open question. CRISOT demonstrates that a model trained on sequence features alone can outperform both learning-based and hypothesis-driven alternatives on independent test sets, achieving an average PR-AUC of 0.46 with statistical significance ($p < 0.05$) [9]. This result matters less for its absolute performance than for what it establishes methodologically: the mapping from guide sequence to off-target propensity is learnable and empirically verifiable, which is precisely the precondition any LLM-based predictor would need. The same paradigm is confirmed by multi-view architectures, where CRISPR-M integrates sequence, structure, and energy views to predict sgRNA off-target effects [51], and by feature-representation studies that treat the prediction task as one of learning discriminative encodings under severe class imbalance [41]. Taken together, these works indicate that the input-output contract for sequence-to-risk prediction is already well specified; the open question concerns not whether the mapping exists but how confidently it can be asserted.

Language-model pretraining has entered this space directly, providing the closest existing analogue to the transfer proposed in this chapter. CCLMoff incorporates an RNA language model pretrained on RNAcentral and demonstrates strong cross-dataset generalization across thirteen genome-wide off-target detection technologies [42]. This is a consequential finding for the transfer argument: it shows that representations learned from large unlabeled sequence corpora carry over to the off-target task, and that generalization across heterogeneous measurement platforms is achievable rather than merely aspirational. The mechanism is intuitive—pretraining captures nucleotide-level regularities that supervised datasets are too small and too platform-specific to supply—but its significance is architectural, because it establishes that a language-model backbone can serve as the representational substrate for CRISPR risk prediction.

A parallel result appears in the DNA foundation model setting, where DNABERT-Epi reports a PR-AUC improvement of +0.1653 ($p < 0.001$) over models trained from scratch, with epigenomic features further improving F1, MCC, and PR-AUC at comparable significance [10]. The convergence of RNA- and DNA-based evidence suggests that pretraining benefit is a property of the modality, not of a particular architecture.

However, the immune-risk side of the question presents a markedly different picture, and the contrast is instructive. Where off-target prediction has absorbed language-model methods, immunogenicity prediction remains the domain of structured computational immunology. NetMHCIIpan 4.0 and related algorithms classify 107 monoclonal antibodies into high- and low-immunogenicity groups on the basis of peptide–HLA-II binding affinity, but they depend on linear epitope assumptions and predict imprecisely for novel sequences. The limitation is mechanistic rather than incidental: binding-affinity models encode a biophysical pathway that language models would have to learn implicitly from data that is scarce and assay-dependent. Consequently, the transfer argument is asymmetric. For off-target risk, the precedent is direct and the representational machinery already exists; for immune risk, the input-output relation is mediated by structural biology that current sequence models do not obviously capture, and no study in the assembled evidence attempts the LLM route.

The one architectural precedent that spans both domains is CRISPR-GPT, which operates as an LLM agent and explicitly enumerates off-target prediction among six meta-tasks, dedicating a distinct Off-target Prediction Mode to it. This is the closest existing instance of the "LLM orchestration plus domain predictor" pattern that the transfer would require, and it demonstrates feasibility at the systems level. Yet the precedent is weaker than it first appears. CRISPR-GPT orchestrates external predictors rather than estimating risk from its own representations, so it does not test whether LLM uncertainty is meaningful for the prediction task; and no documented relation in the literature connects it to the LLM safety evaluation cluster. The architecture is therefore transferable while the evaluation is not—a distinction that matters greatly for what follows.

The absence of cross-cluster precedent is itself a finding, and it is best characterized precisely. The AI safety benchmark literature, exemplified by work on how safety benchmarks should benchmark safety, extends outward only to other LLM safety studies—risk assessment frameworks for mental health support, patient-safety evaluation in generative AI development, and conceptual surveys of evaluation-safety failures [50]. No extension, comparison, or support relation reaches the CRISPR off-target or immunogenicity literature. This is not an artifact of insufficient searching but a genuine structural gap: the two clusters have developed independent vocabularies, benchmarks, and failure taxonomies, and neither cites the other's evaluation apparatus. The consequence for the transfer is that it must be argued constructively rather than by analogy to an existing bridge.

What makes the gap bridgeable is a formal isomorphism between the two clusters' evaluation pathologies. The safety benchmark analysis reports that 81% ($N = 170$) of benchmarks evaluate only known-knowns and that 79% ($N = 166$) treat empirical frequencies as probabilities without confidence intervals [50]. The CRISPR side exhibits the same two failures in different dress: off-target predictors confront roughly 97% class imbalance, which forces reliance on ranking metrics such as AUPR while leaving absolute risk uncalibrated [41], and cross-dataset generalization remains weak even for strong models [51]. The correspondence is not superficial. In both cases, the reported quantity is a discriminative score whose relationship to a calibrated probability of harm is left unstated, and in both cases distribution shift across evaluation contexts is acknowledged but not measured. This isomorphism is the strongest available justification for migrating uncertainty machinery across domains.

A sharper contrast emerges when calibration is examined directly. The safety benchmark literature proposes probability–severity formulations and explicit calibration requirements as minimum conditions for a benchmark to report risk meaningfully [50]. Across the CRISPR prediction studies assembled here, no implementation of confidence intervals, post-hoc calibration, or distribution-shift assessment appears; the closest approximation is the deliberate choice of AUPR in response to class imbalance [41], which addresses ranking robustness but not probabilistic validity. The asymmetry is stark: the LLM safety cluster has formalized what a trustworthy risk number requires, while the CRISPR cluster has optimized the discrimination metric without adopting that formalization. An LLM-based CRISPR predictor would inherit the second without the first unless the transfer is made explicit.

The dose and vector dimensions of the question remain the least supported, and honesty about this boundary is essential. Immunogenicity evidence at the therapeutic level is available—engineered low-immunogenicity lines have been characterized immunologically and mutational profiles mapped, and delivery-approach reviews catalog the strengths and limitations of viral and non-viral vectors —but these are descriptive and mechanistic contributions, not predictive models taking vector or dose as inputs.

Similarly, mitigation strategies for in vivo editing and engineered base editors with reduced off-target activity supply design-level knowledge rather than learned risk functions [8]. The literature therefore supports sequence-to-off-target prediction strongly, sequence-to-immunogenicity prediction weakly and only through structural immunology, and vector/dose-to-risk prediction not at all through LLM methods.

This pattern of coverage and blind spots should be read against the corpus as a whole. As the annual distribution of the assembled literature shows, research activity relevant to these questions grows markedly after 2024, and the source distribution indicates that the safety-benchmark and genome-editing literatures occupy largely disjoint venues—a bibliometric shadow of the structural gap identified above. The sub-question coverage map reinforces the same reading: questions concerning predictive modeling of off-target risk are well populated, whereas those concerning calibrated immune-risk prediction under LLM uncertainty are sparsely evidenced. These observations are consistent with, rather than independent of, the citation-level finding that no cross-cluster relation exists.

Synthesizing across these strands, the answer to the section's question is conditional. An LLM can plausibly predict off-target risk from sequence, since language-model pretraining has already demonstrated cross-platform generalization on that task and the input-output contract is well established [42, 10]. It cannot yet be said to predict immune risk from sequence, vector, or dose, because the relevant evidence comes from structural immunology with acknowledged limits on novel sequences and because no study attempts the language-model route [52]. And the precedent for connecting LLM safety evaluation to CRISPR risk prediction is architectural rather than empirical: CRISPR-GPT shows the orchestration pattern but not the uncertainty semantics [34], while the safety literature supplies the semantics without the biological application [50]. The transfer is therefore best framed as the construction of a missing bridge between two well-developed but mutually isolated traditions.

This framing carries a direct implication for how uncertainty should be handled in the transferred setting. If the CRISPR literature has optimized discrimination while leaving probabilistic validity unaddressed, then importing an LLM's uncertainty estimates without a calibration contract would compound rather than resolve the deficiency—the model would supply confidence numbers whose reliability is exactly what the receiving domain has never measured. The next section therefore takes up the question of what such a contract requires, examining how predictive uncertainty can be rendered comparable across measurement platforms and expressed as safety margins that editing protocols can act upon.

9.3 Separating Aleatoric from Epistemic Uncertainty and Calibrating Decision Thresholds for Regulatory Communication

How should the two fundamentally different kinds of uncertainty in CRISPR safety assessment be told apart, propagated, and converted into calibrated thresholds that regulators can act on? The preceding discussion established that confidence values must be made comparable across measurement platforms before they can function as a contract; the present section argues that such a contract is only meaningful once the irreducible stochastic component of editing outcomes is separated from the reducible component attributable to incomplete knowledge, because the two demand opposite responses from both modelers and regulators.

Aleatoric uncertainty in CRISPR safety assessment arises from genuine biological stochasticity, and it therefore sets a floor on achievable predictive precision rather than signaling a deficiency in the model. Off-target profiles vary with cell type, chromatin state, and repair-pathway activity in ways that no predictor can fully resolve, a variability documented across the multi-view and language-model-based prediction approaches that dominate the recent literature [51, 10]. The genome-wide prediction and optimization study makes this floor visible by showing that even comprehensive candidate enumeration leaves a residual class of sites whose activity is context-dependent [9]. This residual is not a modeling failure to be engineered away; it is the irreducible spread that editing protocols must absorb through design margins, dose selection, and monitoring. Treating it as epistemic and continuing to collect data against it wastes resources, whereas treating epistemic gaps as aleatoric prematurely abandons control. The distinction is thus operational before it is statistical.

Epistemic uncertainty, by contrast, is reducible and should be represented as such, because it marks the regions where additional measurement or model refinement would actually change a decision. The versatile off-target prediction tool and the DNABERT-based predictor both exhibit elevated error precisely where training coverage is thin, indicating that their confident predictions and their uncertain ones carry different evidentiary weight [42, 10]. Feature-representation work on explainable off-target prediction reinforces this reading: attribution patterns become unstable for guide sequences unlike those seen during training, a signature of knowledge gaps rather than biological noise [41]. A natural consequence is

that epistemic uncertainty should be propagated as a widening interval that shrinks with evidence, while aleatoric uncertainty should be propagated as a fixed variance term that does not. Collapsing the two into a single confidence score, as most current predictors implicitly do, destroys exactly the information a regulator needs to decide whether further data collection is worthwhile.

The transfer of this separation from LLM uncertainty modeling is not merely analogical; the receiving domain supplies the biological grounding that the source domain lacks. Work on LLM evaluation-safety failures shows that models frequently express high confidence in regions where their competence is untested, and that benchmark design determines whether such miscalibration is detected at all [50]. The safety-integrity framework for automated driving offers a more directly transferable structure, decomposing risk into components that are bounded by design versus components that are bounded by evidence [17]. Applied to editing, this decomposition maps onto aleatoric spread (bounded by protocol design) and epistemic gap (bounded by accumulated measurement). The spectroscopy safety-cage framework makes the mapping concrete by defining an operational range within which model outputs are trusted and outside which they are not, a construction that presupposes a calibrated notion of when uncertainty reflects distance from the training manifold. These three sources converge on the same architectural claim: separation must occur before thresholding, not after.

Once separated, the two components propagate differently through the assessment chain, and this asymmetry is what makes threshold calibration tractable. Epistemic uncertainty compounds across sequential predictions—guide design, off-target scoring, immunogenicity assessment—because each stage inherits the knowledge gaps of its predecessors, a compounding pattern visible in the layered risk assessments reported for therapeutic delivery and immune profiling. Aleatoric uncertainty, being independent across stages under standard assumptions, aggregates by quadrature and therefore grows more slowly. The practical implication is that a pipeline's overall uncertainty is dominated by epistemic terms in exactly the low-coverage regimes where regulatory scrutiny is highest. Reliability-assessment work for machine learning makes the same point normatively: safety arguments must distinguish claims supported by evidence from claims supported only by assumption, and the two require different argumentative burdens [19]. Propagating a single undifferentiated uncertainty therefore overstates confidence in well-covered regions and understates the need for evidence in poorly covered ones.

Threshold calibration then becomes a matter of setting decision boundaries against the aleatoric floor while treating epistemic width as a trigger for escalation rather than a reason for rejection. The algorithm-to-outcome concordance metric provides a template for this two-tier logic by quantifying the gap between predicted and realized clinical benefit and treating large gaps as evidence of translation failure rather than model error [48]. A CRISPR analogue would set an acceptance threshold above the aleatoric floor—since no predictor can be required to resolve biological noise—and a separate escalation threshold on epistemic width, above which a candidate edit is routed to additional empirical validation. The medical-AI security framework supplies the reproducibility discipline needed to make such thresholds auditable across sites, while the patient-safety evaluation work shows that thresholds defined without explicit uncertainty decomposition tend to be either too permissive in deployment or too restrictive in development. Calibration is thus not a single number but a pair of boundaries with different justifications.

Regulatory communication imposes an additional requirement that the technical separation alone does not satisfy: each threshold must be accompanied by an account of what evidence would move it. Benchmarks that report only aggregate safety scores obscure this, as the benchmark-design critique argues when it distinguishes safety measurement from safety communication [50]. Agentic safety probing further demonstrates that deployment-context risk can diverge sharply from benchmark-context risk, so a threshold calibrated on curated data may not transfer to operational settings. The clinical-safety and environmental-impact quantification for therapeutic LLMs illustrates a workable reporting convention: separate point estimates for irreducible and reducible components, each with its own confidence statement. Translating this convention to editing safety would give regulators a document that says not only "this edit is predicted safe" but "this prediction is limited by biological variance to this degree and by knowledge gaps to that degree, and here is what would reduce the latter." Such a statement is what makes the confidence contract actionable rather than decorative.

The boundary conditions of this scheme deserve honest statement, because the separation is cleaner in principle than in current practice. Existing CRISPR predictors rarely report decomposed uncertainty, and the multi-view and language-model architectures that perform best are also the least interpretable in this respect [51, 10]. Where decomposition has been attempted, as in feature-attribution studies, the epistemic component is inferred indirectly from attribution instability rather than measured directly [41]. Off-target mitigation strategies in vivo add a further complication: the relevant uncertainty includes delivery and persistence effects that no sequence-based predictor represents. Base-editing systems with minimized off-

target effects show that protocol design can lower the aleatoric floor itself, meaning the floor is not fixed but is a function of the editing modality chosen [8]. These observations do not undermine the separation; they specify where it must be supplemented by modality-specific evidence before thresholds can be certified.

A second boundary concerns the assumption that aleatoric and epistemic components are independent across the assessment chain, which the compounding analysis above relied upon. Immune and mutational profiling of gene-edited cells suggests that biological variance and knowledge gaps can be correlated—regions of high stochasticity are often also regions of sparse measurement—so quadrature aggregation may underestimate total uncertainty. The immunogenicity-by-design literature makes a similar point from the engineering side: design choices that reduce one risk category can shift variance into another [52]. Where correlation is suspected, the conservative response is to propagate the sum rather than the quadrature combination, accepting wider intervals in exchange for not understating risk. Agentic automation of gene-editing experiments offers a partial remedy, since automated protocols generate the standardized measurements that would let correlation structure be estimated rather than assumed [34]. Until such data exist, threshold calibration should be reported with an explicit statement of the aggregation assumption used.

Taken together, the evidence supports a three-part answer to the question posed at the outset. Aleatoric uncertainty should be estimated as a modality-specific floor, propagated as fixed variance, and used to set the minimum achievable acceptance threshold; epistemic uncertainty should be estimated from coverage and attribution stability, propagated as a widening interval, and used to set an escalation threshold that routes candidates to empirical validation. The two thresholds carry different justifications and must be communicated separately, with an explicit account of what evidence would move each and what aggregation assumption underlies the reported interval. This scheme is currently better specified than it is implemented, and the gap between specification and implementation is itself informative: it locates precisely where CRISPR safety assessment lacks the standardized measurement that would make decomposition routine. Establishing that measurement base is a question of experimental and reporting infrastructure rather than of modeling technique, and it is to the institutional and evidentiary conditions under which uncertainty claims become regulatorily usable that the following section turns.

10 DISCUSSION

The five research questions examined in this review were posed independently, yet their answers converge on two cross-cutting threads. The first is an architectural–task fit principle: the safety profile of a CRISPR intervention is not a property of the nuclease alone but of the pairing between delivery architecture and editing task. Evidence assembled for RQ1 shows that viral vectors, electroporation, biomaterials, and RNP delivery do not each bind a distinct set of safety problems; instead they converge on the same core nodes of off-target effects and immunogenicity, differentiated chiefly by expression duration and intrinsic carrier properties. This finding reframes delivery selection as the primary lever on safety rather than a logistical afterthought, and it recurs in the immunogenicity analysis of RQ2, where vector-driven immune responses are shown to determine editing persistence. The second thread is evaluative fragmentation: the evidence base is rich but partitioned across incompatible measurement regimes. RQ3 documents a star-shaped topology in which only CHANGE-seq and GUIDE-seq attain the status of shared benchmarks, and even these operationalize "active off-target" through incompatible thresholds; RQ4 finds that roughly 60% of documented limitations cluster in a validation-and-deployment, evidence-uncertainty quadrant rather than being distributed across the risk space. Together these threads imply that progress in clinical translation depends less on new editors than on aligning architecture with task and reconciling the scales on which safety is reported.

Because the measurement foundation depends on experimental and reporting infrastructure rather than on modeling technique, whether uncertainty claims can be used at the regulatory level turns instead on the institutional conditions of evidence; and the fact that the five research questions were posed independently yet converge in their answers on two cross-cutting threads indicates precisely that these institutional conditions must ultimately be examined through the concrete pairing of delivery architecture with editing task.

The immunogenicity evidence of RQ2 both supplements and constrains the delivery conclusions of RQ1. RQ1 establishes a gradient of association strength, with viral vectors carrying the most explicit links to immunogenicity and insertion mutagenesis; RQ2 supplies the causal mechanism that explains why this gradient matters, tracing a dose-dependent chain from capsid-specific cytotoxic T-cell responses through transduced-cell clearance to loss of transgene expression, and quantifying the prevalence of pre-existing AAV neutralizing antibodies at 30–50% of the population. This mechanistic account constrains the interpretation of RQ1's "negative or mitigating" classification of biomaterials and RNP delivery: these

approaches are not intrinsically safer in an absolute sense but are safer relative to the specific failure modes that viral persistence amplifies. The constraint runs in the other direction as well. RQ2's finding that pre-existing Cas9 immunity can directly ablate editing in vivo, and that low-immunogenicity variants trade 30–40% of transient expression for reduced antibody titers, means that any efficiency metric reported without an immune-status covariate is incomparable across patient populations. The efficiency–immunogenicity trade-off is therefore not a secondary consideration but a coupling term that any unified objective of the kind envisioned in RQ1 must carry explicitly.

A second cross-question coupling concerns the relationship between measurement fragmentation and computable risk modeling. RQ3's central result—that no shared coordinate axis exists across off-target benchmarks, and that the approximately 97% negative class imbalance inflates accuracy to 0.990–0.992 while AUPR remains at 0.345–0.612—directly limits what RQ4 can achieve. The probabilistic machinery that RQ4 identifies, including probability density assignment with Monte Carlo simulation and P95 percentile ranking, and the hazard-to-probability decomposition borrowed from engineering safety analysis, presupposes commensurable input variables. Where benchmarks disagree on the definition of a positive event, the resulting risk estimates inherit that disagreement as unmodeled variance. Conversely, RQ4 sharpens RQ3's diagnosis by showing that most limitations are not hard biological constraints but estimable distributional uncertainties, which suggests that the absence of a normalization anchor is an institutional rather than a physical obstacle. The consensus across RQ3 and RQ4 is that the field's quantitative apparatus is mature enough to accept structured inputs; the contradiction is that the inputs themselves remain incommensurable.

The transferability analysis of RQ5 exposes both the promise and the limit of importing methods across domains, and in doing so it ties the fragmentation thread to the modeling thread. The aleatoric–epistemic distinction maps almost losslessly onto CRISPR off-target assessment, separating intrinsic stochasticity in seed-region thermodynamics and chromatin accessibility from ignorance about undetected sites and structural variants; bootstrap and delta-method estimators transfer without modification. The attack-success-rate metric, however, requires semantic substitution and direction reversal, because thresholded off-target rates are insensitive to false negatives, as the gap between AUROC 0.9683 and AUPRC 0.51 demonstrates. Most consequentially, the epistemological principle that aggregate scores are comparative rather than universal safety thresholds transfers with the greatest force precisely because it reveals the deepest structural blind spot: no consensus off-target safety threshold currently exists, and the Casgevy characterization framework functions as a case-specific reference rather than a general standard. This is a genuine cross-question contradiction. RQ4 demonstrates that risk models can be constructed and ranked, while RQ5 shows that the field lacks the threshold against which any such ranking could be judged adequate. The coupling is productive rather than paralyzing: it locates the missing piece not in modeling capability but in evaluative governance.

These cross-question findings yield three reusable contributions. First, a classification framework that reorganizes delivery methods not by mechanism but by their position along an expression-duration axis and their association strength with the shared safety nodes of off-target effects and immunogenicity, converting a descriptive taxonomy into a predictive one. Second, a coordinate system for limitations defined by lifecycle stage crossed with hazard type, which separates hard biological constraints that enter a model as priors from estimable uncertainties that enter as distributions, and which makes explicit that the majority of documented limitations are of the latter kind. Third, a design dimension for cross-domain method transfer that grades mappings by their fidelity—lossless for uncertainty decomposition and resampling estimators, semantics-reversing for threshold-count metrics, and epistemologically foundational for the comparative-not-universal principle. Taken together, these contributions reframe the central problem of the field: the binding constraint on credible clinical translation is not the absence of capable editing architectures, nor the absence of modeling formalisms, but the absence of a shared evaluative scale on which architectural choices, immunological trade-offs, and predicted risks can be jointly compared. Establishing that scale is the precondition for converting the assembled evidence into reproducible preclinical-to-clinical decisions.

11 FUTURE RESEARCH

Future Research

Since a comparable evaluation scale has not yet been established, architectural choices, immune trade-offs, and risk prediction cannot be weighed within a single framework; the direction of future research thus shifts from evidence compilation to scale construction, and further asks what methodological and institutional conditions such a scale must satisfy in order to genuinely support reproducible preclinical decision-making.

A balanced reading of the current evidence reveals that the most consequential gaps sit at the interface

between dosing, immunology, and delivery rather than within any single component. The DMD literature establishes that systemic correction demands vector burdens of 5.5×10^{14}–1.8×10^{15} vg/kg, a range that has been reported to overlap with hepatic and renal toxicity in animals and with cardiopulmonary and haematological adverse events in patients, while pre-existing neutralising antibodies block administration in a substantial fraction of recipients (reported seroprevalence spanning 2–60% across capsid variants and populations). No single source in this corpus reports this dose–toxicity–serostatus overlap jointly; the overlap is therefore inferred by juxtaposing these separately sourced findings. Each of these findings is individually credible, yet no study in this corpus reports dose–serostatus–outcome jointly, so the field lacks the stratified evidence needed to define an eligibility envelope. This is an opportunity as much as a deficit: a prospective registry that records vector genome copies, capsid serotype, baseline NAb titres, and organ-function trajectories would convert three separate literatures into one decision-relevant evidence base.

The immunological barrier deserves particular scrutiny because the available claims rest on heterogeneous seroprevalence estimates spanning 2–60% across capsid variants and populations. Cross-reactivity among NAbs and the memory response that forecloses re-dosing are consistently reported, including the observation in non-human primates that AAV8 re-administration raised anti-AAV antibodies without improving transgene expression. What remains missing is longitudinal human data linking titre thresholds to actual transduction failure, and interventional work on transient immunomodulation or capsid switching. Reducing nuclease immunogenicity is an active direction, and multidimensional capsid and regulatory engineering offers plausible routes past the humoral wall, but neither has been evaluated against the dose ceilings that DMD demands.

Off-target prediction illustrates most sharply where strong conclusions outrun the evidence. Reported accuracies near 0.99 coexist with precision as low as 0.009 on K-mer/GUIDE-seq benchmarks, and leading language-model predictors achieve balanced accuracy of 0.998 and AUROC of 0.985 while returning F1 of 0.409 and AUPRC of 0.524. These figures show that aggregate metrics can mask near-useless performance on the rare positive class that matters clinically, a pattern echoed in broader critiques of off-target effects. Sequence-based prediction is nonetheless an empirically validated supervised-learning paradigm, with CRISOT reaching mean PR-AUC of 0.46 and outperforming both learned and hypothesis-driven baselines, and with genome-wide optimisation and accessible tooling already demonstrated. The gap is therefore not feasibility but evaluation practice: precision-recall reporting on clinically weighted positives, external validation across detection assays, and explicit calibration are the missing standards.

Chromatin context supplies the most promising lever for closing that gap, because it is both mechanistically motivated and empirically supported. GUIDE-seq off-target sites are significantly enriched in open chromatin as measured by ATAC-seq, and SHAP attribution for DNABERT-Epi identifies H3K27ac as a dominant predictor of intracellular off-target activity. Epigenome-aware models thus connect the formal probabilistic machinery of Bayesian networks, with its directed acyclic graph and conditional dependencies, to measurable biological state rather than to sequence alone. The critical caveat is that these associations are correlational and drawn from a small number of cell types; without perturbation experiments that hold sequence constant and vary chromatin, the causal claim remains underdetermined. Extending explainable feature representations to chromatin-informed inputs is a tractable next step.

Synthesising across these strands, one leverage point recurs: the absence of shared, clinically anchored evaluation standards that bind delivery, immunology, and prediction into a single evidence chain. Delivery research for neurological and systemic indications documents diverse strategies whose comparative performance is rarely benchmarked on common endpoints, and clinical translation reviews repeatedly call for harmonised reporting. Therapeutic CRISPR delivery analyses reach the same conclusion from the engineering side, while mitigation studies note that off-target evaluation itself lacks agreed thresholds. A research agenda that prioritises (i) stratified dose–serostatus cohorts, (ii) precision-recall and calibration standards for off-target models, and (iii) chromatin-perturbation validation would address several deficits with one investment, and would let immunogenicity-reduction and capsid-engineering efforts be judged against endpoints that reflect actual clinical constraints rather than surrogate aggregates.

12 LIMITATIONS

The evidence base combines an automated retrieval pool with manual supplementation, and this combination bounds what the review can claim to cover. The automated searches drew on a fixed set of databases and query strings, so literature indexed outside those sources, described with non-standard terminology, or published in venues not captured by the queries is likely underrepresented; manual supplementation partially offsets this gap but follows the authors' own judgment of relevance and therefore inherits its biases. Retrieval closed at a predetermined cutoff date, and work appearing after that date is excluded

regardless of its bearing on the review's conclusions.

Methodologically, the review invests in verifiability. Each claim is annotated with the literature that supports it, and the analytical process is documented so that coding decisions and intermediate steps can be traced and audited. These measures make the reasoning inspectable, but they do not eliminate error: individual extraction inaccuracies may persist in the reported evidence, and readers should treat specific details as pointers to the underlying sources rather than as authoritative restatements.

In scope, the review concentrates on its defined topic and admits adjacent fields only where they intersect that topic directly. This focus keeps the synthesis coherent but necessarily excludes material whose relevance is indirect, and findings from neighboring areas appear here only to the extent that they bear on the central question. The boundaries above define the review's intended reach; claims beyond them fall outside what the assembled evidence can support.

13 CONCLUSION

This review set out to determine whether the available evidence supports a unified, quantitative treatment of CRISPR delivery efficiency and editing safety. Across the four thematic strands, the answer is consistent and conditional. On delivery, the reviewed studies document explicit and implicit associations between viral vectors, electroporation, biomaterials-mediated delivery, and RNP delivery on the one hand, and safety-relevant outcomes on the other, yet no study formalizes delivery efficiency and editing safety (off-target plus immunogenicity) into a single comparable scalar. On immunogenicity, the causal chain from vector immunogenicity through immune response and editing persistence to clinical risk is largely established, but it has not been integrated into a unified risk model, and the strength of extractable contrastive evidence varies sharply across the four vector classes. On off-target benchmarking, the reviewed studies form a star-shaped comparison structure with sparse cross-benchmark links and no genuine normalization anchor, so only protocol-bound, conditional risk scales—not absolute cross-laboratory scales—can be defined. On translation, the reported limitations concentrate on a few structural dimensions, and roadmap papers supply structurable risk elements that map only partially onto the risk-matrix dimensions.

Now that the boundaries have been delineated, the scope of evidence coverage becomes clear accordingly; however, whether a unified and quantitative conclusion can be drawn within these boundaries still depends on the form of evidence presented by each of the four thematic threads.

Taken together, these findings mark a shift in how the field should read its own evidence: the binding constraint on quantitative CRISPR safety is not a shortage of mechanistic findings but the absence of shared, calibrated anchors that would make heterogeneous results commensurable. The evidence already contains the causal ingredients; what it lacks is the connective framework that turns them into comparable quantities.

For researchers, the practical implication is to treat benchmark normalization and uncertainty decomposition as first-class deliverables rather than post hoc reporting choices, and to attach explicit protocol and prior conditions to every reported safety metric. For engineers, the implication is to design delivery and dose decisions against conditional, context-bound risk scales, and to record the assumptions and constraints under which those scales were derived so that they remain traceable and reusable across systems.

REFERENCES

[1] Jia Yi Goh, Shaun Khoo, Nyx Iskandar, Gabriel Chua, Leanne Tan, and Jessica Foo. Measuring what matters: A framework for evaluating safety risks in real-world llm applications. Preprint.

[2] Alireza Abbaszadeh and Armita Shahlai. Artificial intelligence for crispr guide rna design: Explainable models and off-target safety. Preprint.

[3] Shirui Wang, Zhihui Tang, Huaxia Yang, Qiuhong Gong, Tiantian Gu, Hongyang Ma, Yongxin Wang, Wubin Sun, Zeliang Lian, Kehang Mao, Yinan Jiang, and Zhicheng Huang. A novel evaluation benchmark for medical llms: Illuminating safety and effectiveness in clinical domains. Preprint.

[4] Zhidu Song, Ying Tao, Yue Liu, and Jian Li. Advances in delivery systems for crispr/cas-mediated cancer treatment: a focus on viral vectors and extracellular vesicles, 2026. Preprint.

[5] Jonathan O'Keeffe Ahern, Irene Lara-Sáez, Dezhong Zhou, Rodolfo Murillas, José Bonafont, Ángeles Mencía, Marta García, Darío Manzanares, Jennifer Lynch, Ruth Foley, Qian Xu, and A Sigen. Non-viral delivery of crispr–cas9 complexes for targeted gene editing via a polymer delivery system, 2026. Preprint.

[6] Xiaowen Huang, Aifang Li, Peng Xu, Yangfan Yu, Shuxuan Li, Lina Hu, and Shuying Feng. Current and prospective strategies for advancing the targeted delivery of crispr/cas system via extracellular vesicles, 2026. Preprint.

[7] Menno Hoekstra and Miranda Van Eck. Gene editing for the treatment of hypercholesterolemia, 2026. Preprint.

[8] Ang Li, Hitoshi Mitsunobu, Shin Yoshioka, T. Suzuki, Akihiko Kondo, and Keiji Nishida. Cytosine base editing systems with minimized off-target effect and molecular size, 2026. Preprint.

[9] Qinchang Chen, Guohui Chuai, Haihang Zhang, Jin Tang, Liwen Duan, Huan Guan, Wenhui Li, Wannian Li, Jiaying Wen, Erwei Zuo, Qing Hua Zhang, and Qi Liu. Genome-wide crispr off-target prediction and optimization using rna-dna interaction fingerprints, 2026. Preprint.

[10] Kai Kimata and Kenji Satou. Improved crispr/cas9 off-target prediction with dnabert and epigenetic features, 2026. Preprint.

[11] Jingze Liu and Jiahao Ma. Comparative analysis of machine learning algorithms for predicting on-target and off-target effects of crispr-cas13d for gene editing. Preprint.

[12] Zuben E. Sauna, Vibha Jawa, Sathy V. Balu-Iyer, and Narendra B. Chirmule. Understanding pre-clinical and clinical immunogenicity risks in novel biotherapeutics development. Preprint.

[13] Lishuang Xia, Jia Qin, and Sheng Ye. Addressing the challenges of immunogenicity, heterogeneity and tumorigenicity in the clinical translation of human embryonic stem cells, 2026. Preprint.

[14] Radwa Ewaisha and Karen S. Anderson. Immunogenicity of crispr therapeutics—critical considerations for clinical translation, 2026. Preprint.

[15] Tahereh Mohammadian Gol, Guillermo Ureña-Bailén, Yujuan Hou, Ralph Sinn, Justin S. Antony, Rupert Handgretinger, and Markus Mezger. Crispr medicine for blood disorders: Progress and challenges in delivery, 2026. Preprint.

[16] Tian Zhang, Wenshan Guan, Hao Miao, Xiujie Huang, Zhiquan Liu, Chaonan Wang, Quanlong Guan, Liangda Fang, and Zhifei Duan. Vsrq: Quantitative assessment method for safety risk of vehicle intelligent connected system, 2026. Preprint.

[17] Moritz Werling, Rainer Faller, Wolfgang Betz, and Daniel Straub. Safety integrity framework for automated driving, 2026. Preprint.

[18] Daniel Betschinske, Malte Schrimpf, Steven Peters, Kamil Klonecki, Jan Peter Karch, and Moritz Lippert. Towards more efficient quantitative safety validation of residual risk for assisted and automated driving. Preprint.

[19] Reliability assessment and safety arguments for machine learning, 2026. Preprint.

[20] Florian Geissler, Syed Qutub, Sayanta Roychowdhury, Ali Asgari, Yang Peng, Akash Dhamasia, Ralf Graefe, Karthik Pattabiraman, and Michael Paulitsch. Towards a safety case for hardware fault tolerance in convolutional neural networks using activation range supervision. Preprint.

[21] Fernando Acebes, José Manuel González-Varona, Adolfo López-Paredes, and Javier Pajares. Beyond probability-impact matrices in project risk management: A quantitative methodology for risk prioritisation, 2026. Preprint.

[22] Kehao Zhu, Yingye Zheng, and Kwun Chuen Gary Chan. Weighted brier score - an overall summary measure for risk prediction models with clinical utility consideration, 2026. Preprint.

[23] Angel Paul, Dhivin Shaji, Lifeng Han, Warren Del-Pinto, Goran Nenadic, and Suzan Verberne. Deid-clinic: A risk-aware pseudonymization framework for clinical text de-identification and re-identification risk assessment. Preprint.

[24] Shaona Ghosh, Prasoon Varshney, Makesh Narsimhan Sreedhar, Aishwarya Padmakumar, Traian Rebedea, Jibin Rajan Varghese, and Christopher Parisien. Aegis2.0: A diverse ai safety dataset and risks taxonomy for alignment of llm guardrails. Preprint.

[25] Junfeng Jiao, Saleh Afroogh, Kevin Chen, Abhejay Murali, David Atkinson, and Amit Dhurandhar. Llms and childhood safety: Identifying risks and proposing a protection framework for safe child-llm interaction. Preprint.

[26] Gandlf: the generally nuanced deep learning framework for scalable end-to-end clinical workflows, 2026. Preprint.

[27] Chengxin Zhang, Huizhong Guo, Zifei Wang, Fred Feng, Anuj Pradhan, and Shan Bao. Assessing the effectiveness of driver training interventions in improving safe engagement with vehicle automation systems, 2026. Preprint.

[28] Zilong Xu, Zihao Wang, He Li, Dingli Yu, Zaili Yang, and Jin Wang. A geometric analysis-based safety assessment framework for marine vehicle route decision-making, 2026. Preprint.

[29] Dachuan Zuo, Zilin Bian, Fan Zuo, and Kaan Ozbay. Composite safety potential field for highway driving risk assessment. Preprint.

[30] A terminology and quantitative framework for assessing the habitability of planets and moons, 2026. Preprint.

[31] Maryam Rezaee, Farideh Izadi, Saeed Nobaharian, Leila Hassani, Esmaeil Rezaei, Sara Sadeghi Amini, Ali Akbar Pourfathollah, Ali Kowsari, Sirous Naeimi, Khalil Khashei Varnamkhasti, and Leila Rouhi. Therapeutic crispr/cas9 delivery approaches: strengths, limitations, and a roadmap for clinical translation, 2026. Preprint.

[32] Motahareh Arjomandnejad, Ishani Dasgupta, Terence R. Flotte, and Allison M. Keeler. Immunogenicity of recombinant adeno-associated virus (aav) vectors for gene transfer, 2026. Preprint.

[33] Anuradha Bhardwaj. Navigating off-target effects in crispr-based genome editing for safer gene therapies, 2026. Preprint.

[34] Yuanhao Qu, Kaixuan Huang, Ming Yin, Kanghong Zhan, Dyllan Liu, Di Yin, Henry C. Cousins, William A. Johnson, Xiaotong Wang, Mihir Shah, Russ B. Altman, and Denny Zhou. Crispr-gpt for agentic automation of gene-editing experiments, 2026. Preprint.

[35] Bilge DEBELEÇ BÜTÜNER. Overview of delivery methods for gene editing, 2026. Preprint.

[36] Congting Guo, Xiaoteng Ma, Fei Gao, and Yuxuan Guo. Off-target effects in crispr/cas9 gene editing, 2026. Preprint.

[37] Jad Madani, Humza N. Zubair, Ghazi-Abdullah Saroya, Rachel E.N. Reyes, and Rami Madani. Comparing crispr-cas9 gene therapy delivery systems: Implications and optimization for hyperargininemia, 2026. Preprint.

[38] Ankit Kumar Dubey and Ebrahim Mostafavi. Biomaterials-mediated crispr/cas9 delivery: recent challenges and opportunities in gene therapy, 2026. Preprint.

[39] Xingqi Chen. Research on reducing the immunogenicity of nucleases in gene therapy, 2026. Preprint.

[40] Leonardo Martin, Jure Bohinc, Alessandra Recchia, Stefano Gritti, Giorgia Santilli, Joanna Zeyland, Melita Vidaković, Nevena Grdović, Karim Benabdellah, María Ortiz-Bueno, Bilge DEBELEÇ BÜTÜNER, and Luana Pisaniello. In vivo delivery strategies for therapeutic crispr genome editing, 2026. Preprint.

[41] Faiza Hasin, Michele Minervini, Corrado Mencar, Giuseppe Ventrella, Arianna Consiglio, Alessandro Orro, and Tommaso Selmi. Feature representation for explainable crispr off-target prediction and base editing efficiency, 2026. Preprint.

[42] Weian Du, Liang Zhao, Kaichuan Diao, Yangyang Zheng, Qianyong Yang, Zhenzhen Zhu, Xiangxing Zhu, and Dongsheng Tang. A versatile crispr/cas9 system off-target prediction tool using language model, 2026. Preprint.

[43] Yanbing Xu. How to mitigate crispr-cas9 off-target effects: Evaluation of current detection, prediction, and mitigation strategies, 2026. Preprint.

[44] Hossein Sholehrasa, Xuan Xu, Doina Caragea, Jim E. Riviere, and Majid Jaberi-Douraki. Predictive modeling and explainable ai for veterinary safety profiles, residue assessment, and health outcomes using real-world data and..., 2026. Preprint.

[45] He Wen. The digital divide in process safety: Quantitative risk analysis of human-ai collaboration, 2026. Preprint.

[46] Victor Murcia Pienkowski, Tamara Boschert, Piotr Skoczylas, Anna Sanecka-Duin, Maciej Jasiński, Bartłomiej Król-Józaga, Giovanni Mazzocco, Sławomir Stachura, Lukas Bunse, Jan Kaczmarczyk, Edward W. Green, and Agnieszka Blum. Arditox: platform for the prediction of tcrs potential off-target binding, 2026. Preprint.

[47] Nikki Grens, Luís F. Simões, Kai Hou Yip, and Theresa Lueftinger. Operational range bounding in spectroscopy: A safety cage framework for machine learning models, 2026. Preprint.

[48] A proposed framework for quantifying ai-to-clinical translation: The algorithm-to-outcome concordance (aoc) metric, 2026. Preprint.

[49] Jingnan Zheng, Yanzhen Luo, Jingjun Xu, Bingnan Liu, Yuxin Chen, Chenhang Cui, Gelei Deng, Chaochao Lu, Xiang Wang, An Zhang, and Tat-Seng Chua. Risky-bench: Probing agentic safety risks under real-world deployment, 2026. Preprint.

[50] Cheng Yu, Severin Engelmann, Ruoxuan Cao, Dalia Ali, and Orestis Papakyriakopoulos. How should ai safety benchmarks benchmark safety?, 2026. Preprint.

[51] Jialiang Sun, Jun Hai Guo, and Jian Liu. Crispr-m: Predicting sgrna off-target effect using a multi-view deep learning network, 2026. Preprint.

[52] Chantal T. Harris and Sivan Cohen. Reducing immunogenicity by design: Approaches to minimize immunogenicity of monoclonal antibodies, 2026. Preprint.